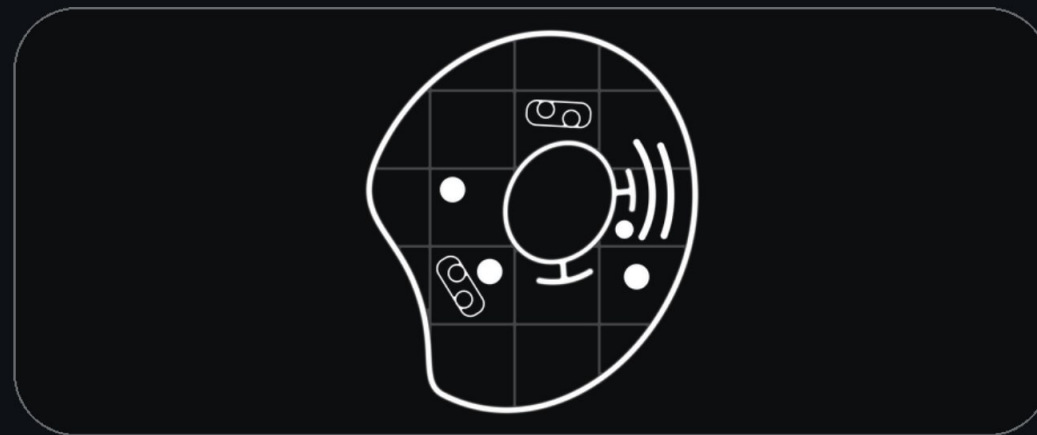
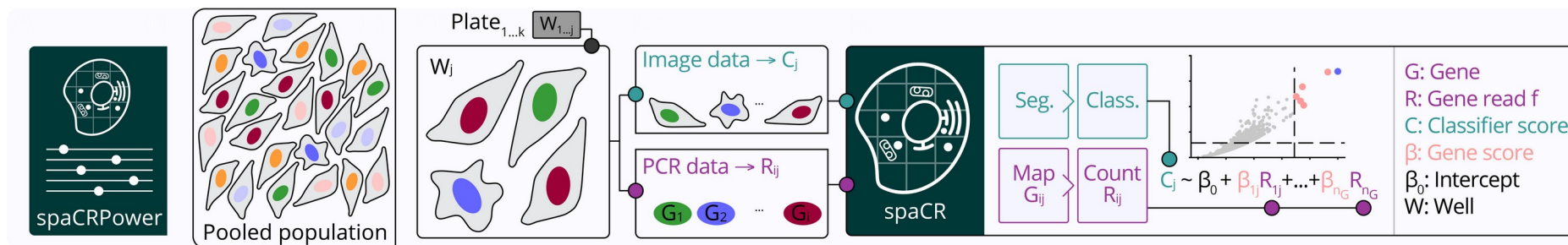


spaCR

Spatial phenotype analysis of CRISPR
screens



What is spaCR?



Pooled screen: every well holds a mix of knockouts. Image it, sequence it, and regress. (Preprint, Fig. 1A)



Image every well

Segment, measure and classify every cell. The phenotype becomes a number per well.



Sequence every well

Read the gRNA barcodes to learn which knockouts are in each well, and in what proportion.



Link the two

Regress the phenotype on guide composition. Out comes an effect per gene: a ranked hit list.

No sequencing? Segmentation, measurement, annotation, classification and the assays work on any high-content imaging experiment.

INTRODUCTION

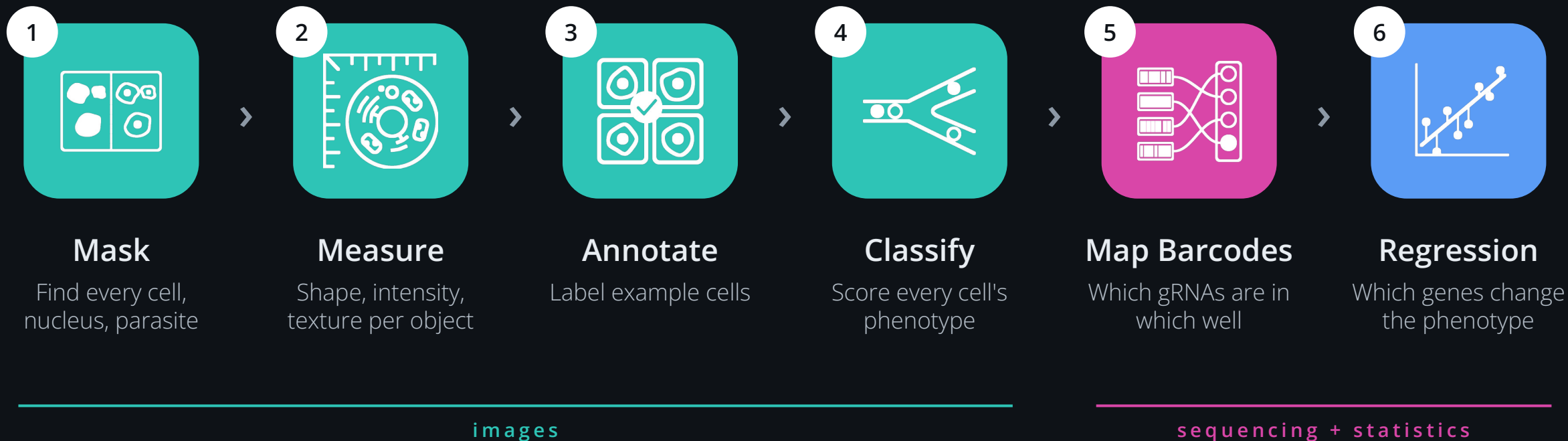
Where spaCR fits among existing tools

- tool lacks it
- tool has it: spaCR has it too
- tool has it: spaCR does not

	CellProfiler	Fiji / ImageJ	napari	Cellpose	ilastik	QuPath	DL4Mic	MAGeCK	spaCR
<i>Best for</i>	<i>cell measurement</i>	<i>image analysis</i>	<i>image viewing</i>	<i>segmentation</i>	<i>pixel / object ML</i>	<i>tissue slides</i>	<i>no-code DL</i>	<i>sequencing screens</i>	<i>image-based screens</i>
Point-and-click interface	✓	✓	✓	✓	✓	✓	limited	✗	✓
Batch and headless runs (server, cluster)	✓	limited	✗	✓	✓	✓	limited	✓	✓
Reads most microscope file formats	✓	✓	plugin	limited	limited	✓	limited	✗	✓
Deep-learning segmentation	plugin	plugin	plugin	✓	limited	plugin	✓	✗	✓
Train your own segmentation model	✗	plugin	plugin	✓	✓	limited	✓	✗	✓
Live assisted segmentation	✗	plugin	plugin	limited	✓	✓	✗	✗	✓
Correct masks by hand	limited	✓	✓	✓	✓	✓	✗	✗	✓
Per-cell measurements	✓	✓	plugin	✗	✓	✓	✗	✗	✓
3-D (z-stack) images	✓	✓	✓	✓	✓	limited	limited	✗	beta
Time-lapse tracking	✓	plugin	plugin	✗	✓	✗	✗	✗	✓
Whole plates in one run	✓	limited	✗	limited	limited	limited	limited	✗	✓
Plate maps, heat maps and QC	limited	✗	✗	✗	✗	✗	✗	✗	✓
Classify cells from images (deep learning)	✗	✗	✗	✗	✗	✗	✗	✗	✓
Classify cells from measurements	limited	✗	plugin	✗	✓	✓	✗	✗	✓
Explore cells: UMAP with images, gating	limited	✗	plugin	✗	✗	✗	✗	✗	✓
Plots and statistics without code	limited	limited	plugin	✗	✗	limited	✗	limited	✓
GPU acceleration	✗	plugin	plugin	limited	limited	limited	✓	✗	✓
Pooled CRISPR screens: images to genes	✗	✗	✗	✗	✗	✗	✗	limited	✓
Pixel classifier from brush strokes	✗	plugin	plugin	✗	✓	✓	✗	✗	✗
Whole-slide / tissue images	✗	limited	✗	✗	✗	✓	✗	✗	✗
Large plugin ecosystem	limited	✓	✓	✗	✗	limited	✗	✗	✗

Core features; 'plugin' = needs an add-on, 'limited' = partly. Simplified: check each tool for details.

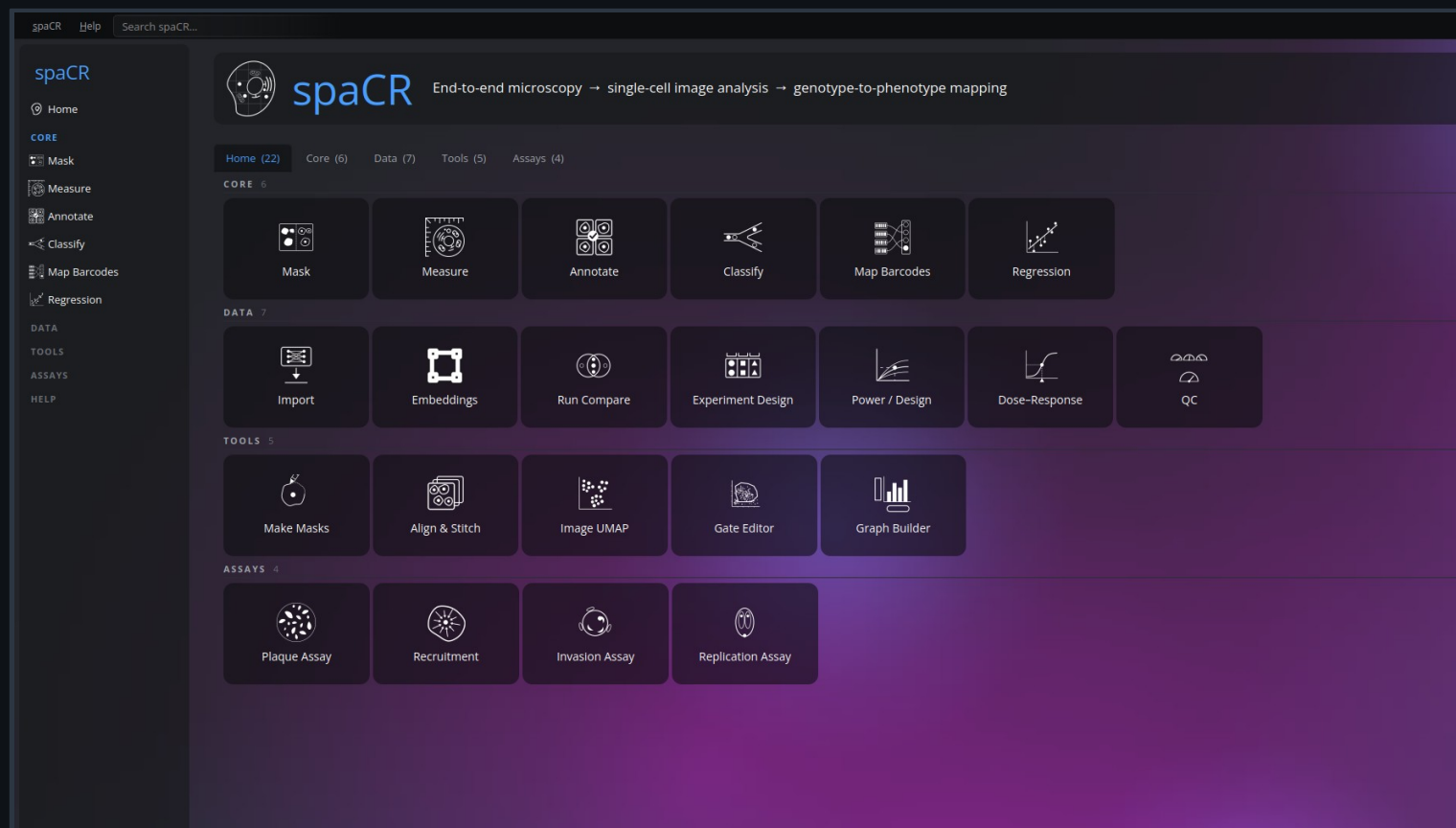
Six steps from plate to hit list



One project, one database. Images, masks, crops, measurements, annotations, predictions, barcodes and well IDs all live in one SQLite file (measurements.db). Every module reads from it and writes back to it, and every run saves its settings so it can be replayed.

INTRODUCTION

22 modules in one desktop app



The home screen: every module is a tile, grouped by what it is for

6

Core

Mask · Measure · Annotate · Classify · Map Barcodes · Regression

7

Data

Import · Embeddings · Run Compare · Experiment Design · Power · Dose-Response · QC

5

Tools

Make Masks · Align & Stitch · Image UMAP · Gate Editor · Graph Builder

4

Assays

Plaque · Recruitment · Invasion · Replication

Plus ~40 sub-tools inside the modules. Ctrl+K opens a command palette that reaches every one of them.

Runs on a laptop, a workstation or a cluster

Desktop app

Point-and-click Qt interface. Live previews, a console per module, settings search, recipes, and a Run button.

Headless

The same modules and settings files run without the window, on a server or a cluster: or send jobs there from the app.

Reproducible

Every run records its settings. spacerepro RUN_DIR replays it. Run History and Run Compare show what changed.

4

kinds of graphics card

NVIDIA, AMD, Apple and Intel: or just the CPU

10

interface languages

switch in Preferences without restarting

77

interactive tutorials

narrated videos from install to hit investigation

3.9–3.14

Python versions

3.12 recommended; Linux, macOS, Windows

INTRODUCTION

Install spaCR

Desktop installer

Easiest. Bundles its own Python: no conda needed.

```
GitHub → Releases → latest
Windows .exe
macOS .pkg (Intel + Apple silicon)
Linux  chmod +x SpaCR-*.run && ./SpaCR-*.run
```

pip (conda env)

Most flexible; add optional extras.

```
conda create -n spacr python=3.12 -y
conda activate spacr
pip install "spacr[qt]"
spacr
```

conda-forge

One command into an existing env.

```
conda create -n spacr python=3.12 -y
conda activate spacr
conda install conda-forge::spacr
spacr
```

From source

For development (nightly branch).

```
git clone --branch nightly \
  https://github.com/EinarOlafsson/spacr
cd spacr && pip install -e .
spacr
```

Something off? `spacr-doctor` checks the install and says how to fix it. Headless server: `pip install spacr` (no Qt).

01

Core pipeline

From raw plate images and FASTQ reads to a ranked list of genes.



Mask



Measure



Annotate



Classify



Map Barcodes



Regression



CORE · 1 OF 6

STABLE

Mask: segmentation

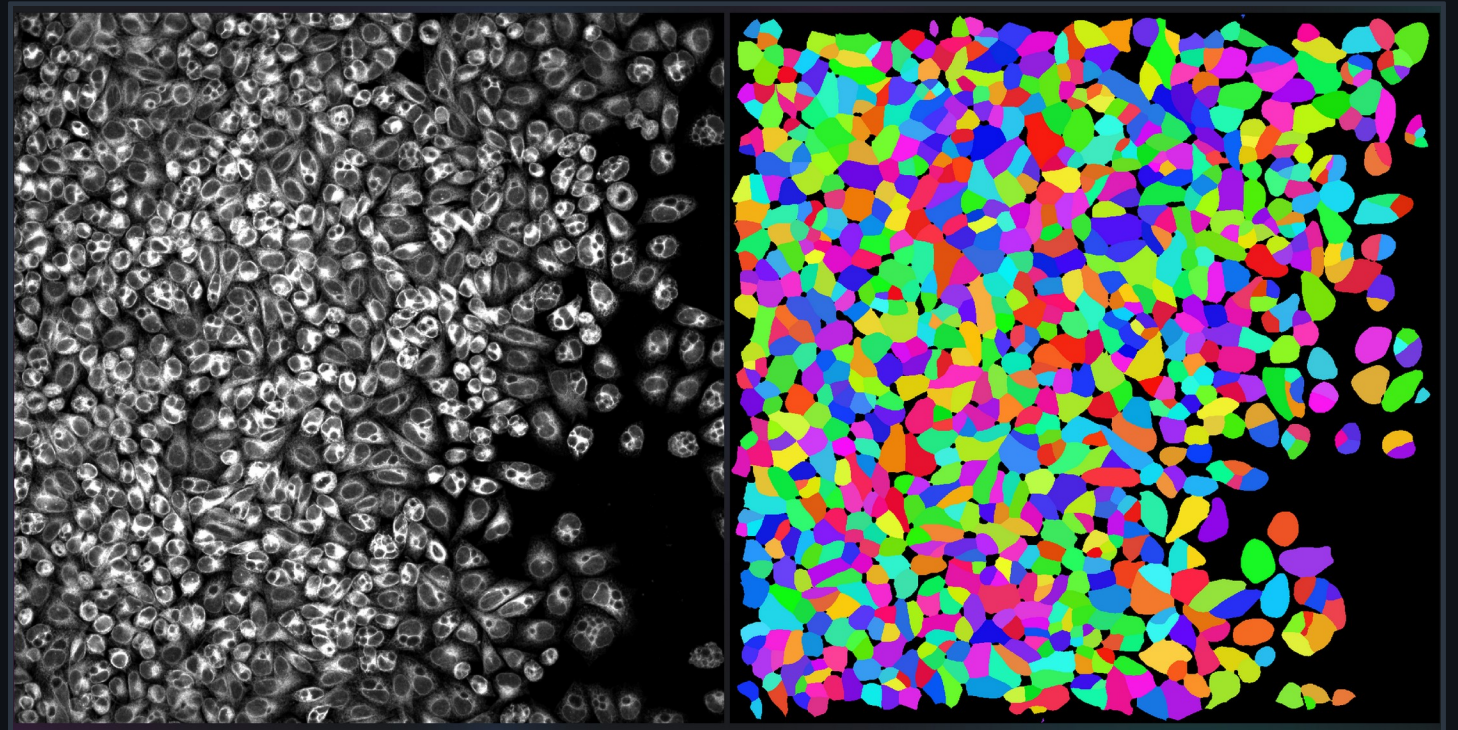
Finds and outlines every cell, nucleus, parasite and organelle in your images.

WHAT YOU CAN DO

- Reads files from most microscopes (TIFF, Leica, Zeiss, Nikon)
- Cellpose-SAM by default; other segmentation models optional
- Cell, nucleus, pathogen and up to 26 organelle classes
- Plug in fine-tuned Model Zoo models (e.g. Toxoplasma vacuoles)
- 2-D, 3-D z-stacks and time-lapse; live preview before a full run

OUTPUT

- One mask per object type, aligned to the images
- Overlay images and a quality score per field
- The settings used, saved with the run



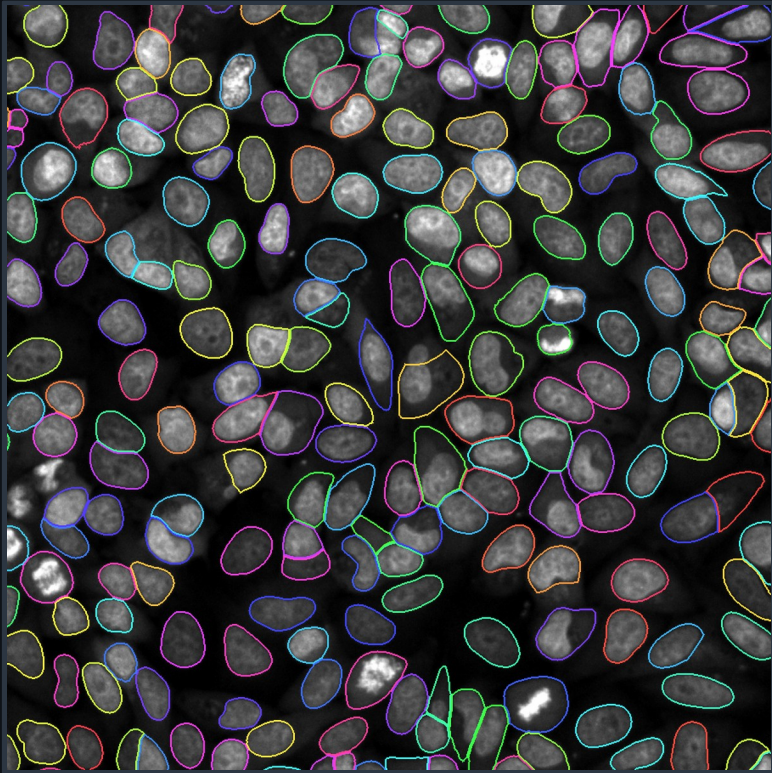
Live preview: host cells (left) and the label mask Cellpose-SAM returns (right)



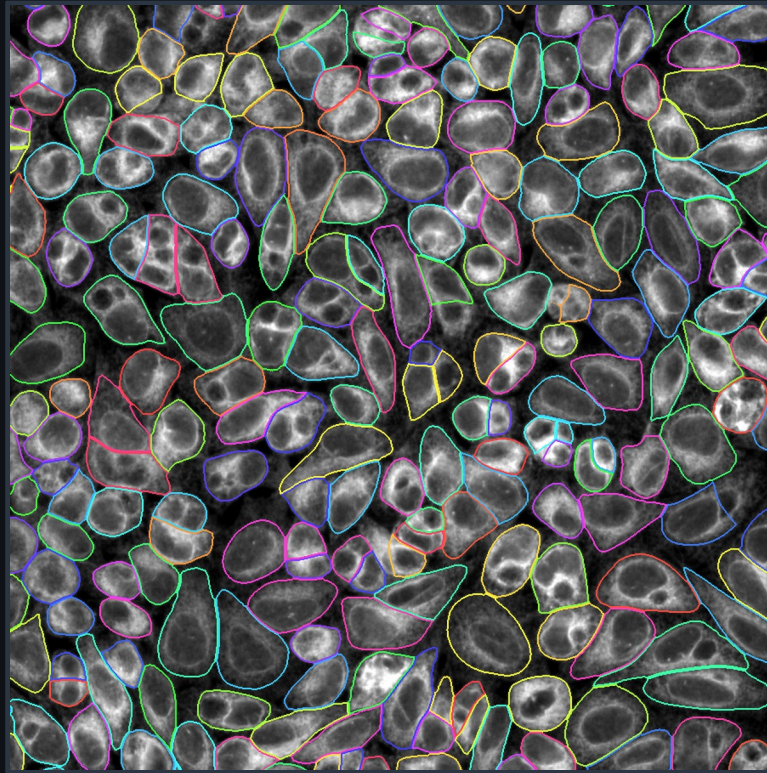
CORE · 1 OF 6

STABLE

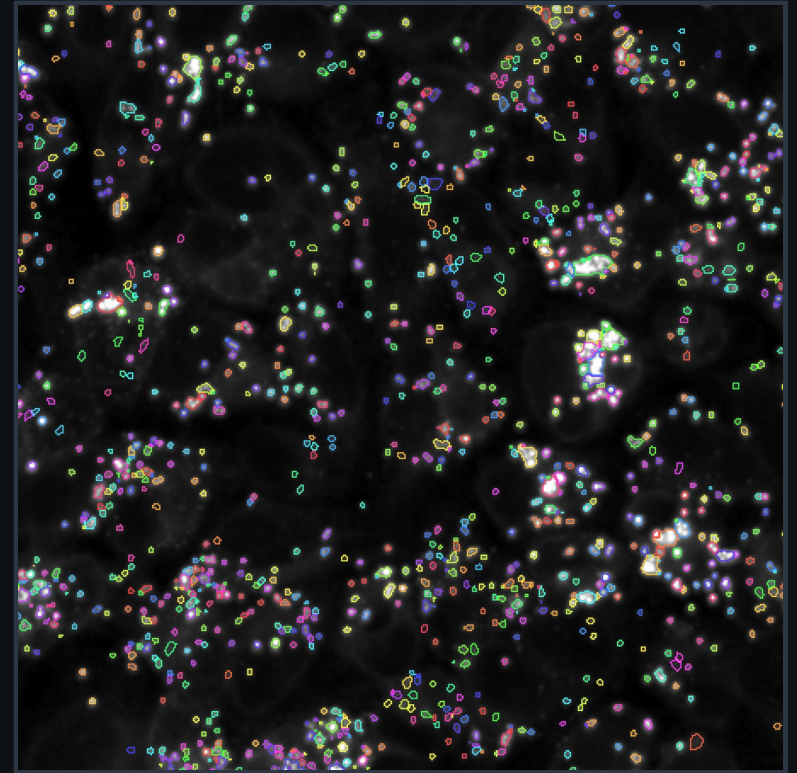
Mask: every object class



Nuclei: Cellpose-SAM on the Hoechst channel



Cells: Cellpose-SAM on the cell stain



Organelles: lipid droplets found as spots (LoG), closer view

SUB-TOOLS INSIDE THIS MODULE

Timelapse

Follow each cell and parasite from frame to frame in movies

OPS

Optical pooled screens: read the barcodes directly in the images (alpha)

Cellpose flows

See why two cells were merged, or one was split



CORE · 2 OF 6

Measure: features and crops

STABLE

Turns every segmented object into a row of numbers: shape, brightness, texture and position.

WHAT YOU CAN DO

- Shape, size, brightness and texture of every object
- Position: neighbours, distances, touching the edge
- Overlap between channels; measure only inside a region you choose
- Saves a crop of every object for Annotate and Classify
- 3-D objects measured in μm

OUTPUT

- One table per object type (cells, nuclei, parasites, ...)
- Per-object PNG crops

SUB-TOOLS INSIDE THIS MODULE

Illumination

Flat-field correction estimated from the plate itself

AnnData export

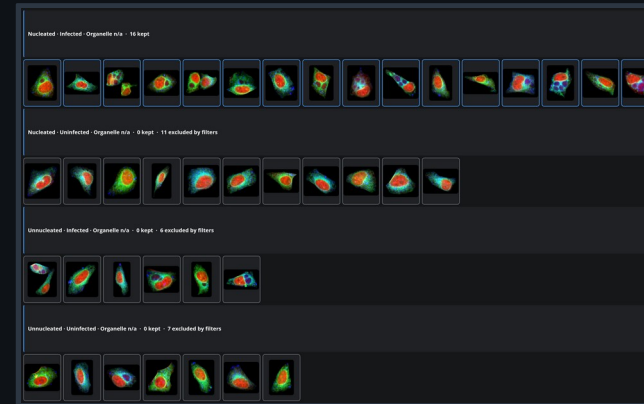
Export for single-cell analysis tools (scanpy)

Motility assay

Track velocity and straightness, split by infection state

Database Browser

Browse, filter and export any project table



Crop preview, grouped by nucleated / infected state

The cell table in the Database Browser



CORE · 3 OF 6

STABLE

Annotate: label cells by hand

Label single-cell crops so a classifier can learn what each phenotype looks like.

WHAT YOU CAN DO

- Pages through a grid of crops from the project
- Click, or press 1–9 with auto-advance; 0 clears, u undoes
- Several annotation columns side by side (e.g. two annotators)
- Active learning: shows the cells the model is least sure about first
- Change channels and colours for display only; files stay untouched

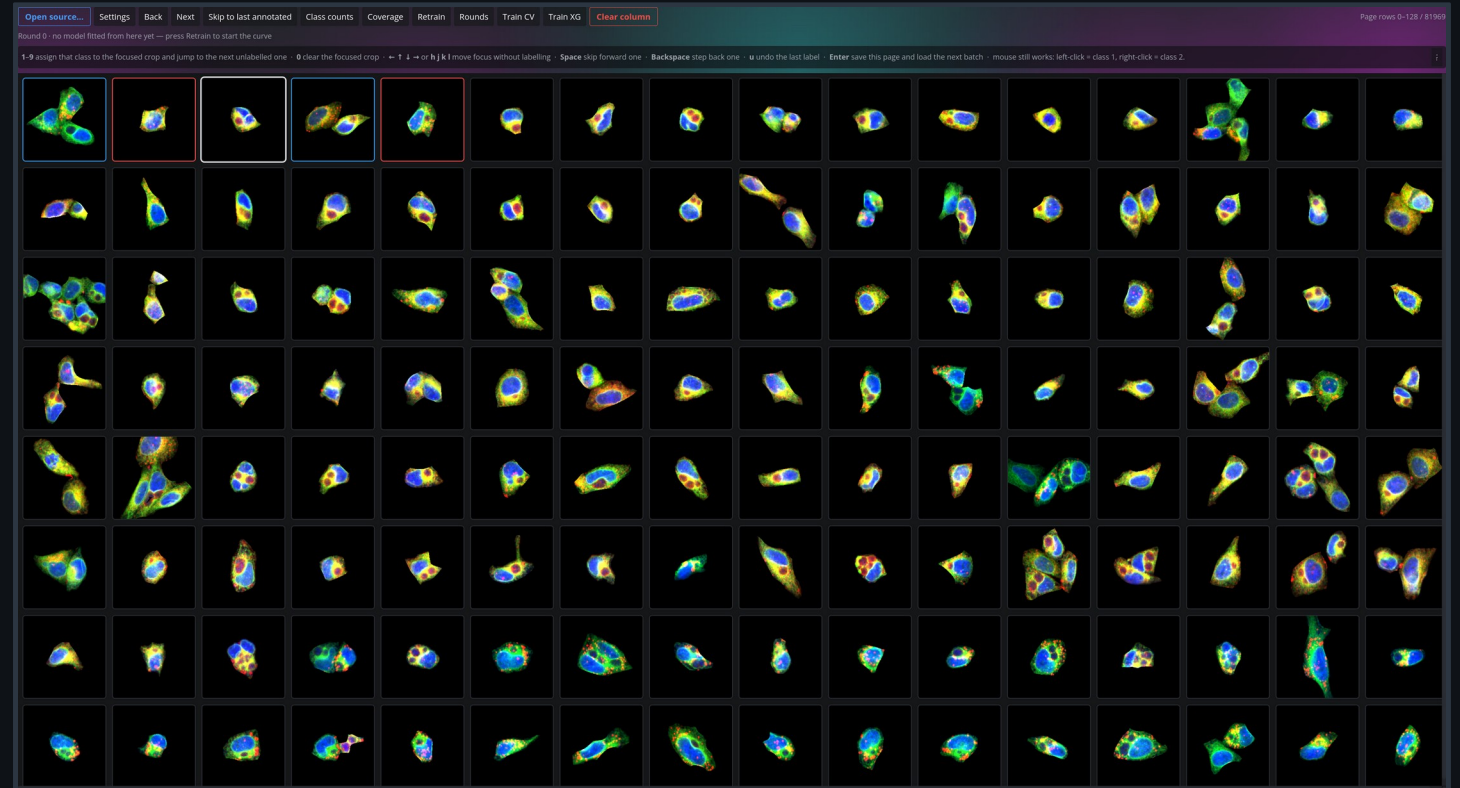
OUTPUT

- Labels in an annotation column of the project
- Class counts and coverage per well

SUB-TOOLS INSIDE THIS MODULE

Annotator Agreement

How often two people label a cell the same way; and a list of the cells they disagreed on



Annotation grid. Selected crops are outlined; the queue can be ranked by model uncertainty



Classify: score every cell

Teach the computer what a phenotype looks like, then let it score every cell in the screen.

WHAT YOU CAN DO

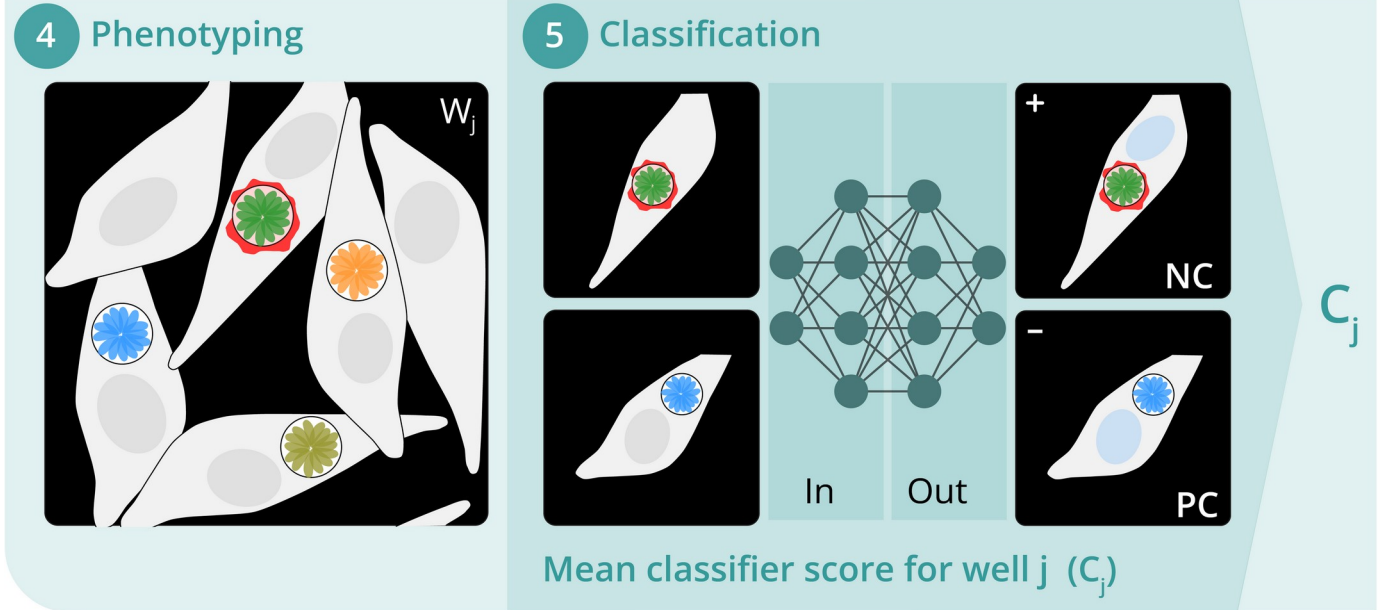
Image model: learns straight from the cell pictures (ResNet, EfficientNet, ViT, MaxViT, ...)

Measurement model: learns from the measured features (XGBoost, random forest, ...)

- Examples come from control wells or from your Annotate labels
- Tested on wells it never trained on, so the accuracy is honest
- Scores every cell and writes the result back to the project

OUTPUT

- A class and a probability for every cell
- The mean score per well → input to Regression



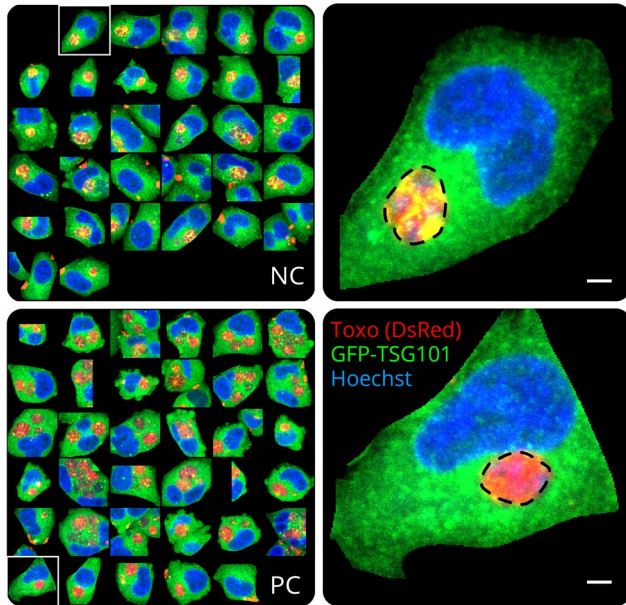
Each cell is scored by the classifier; a well's score is the mean of its cells (preprint Fig. 2)



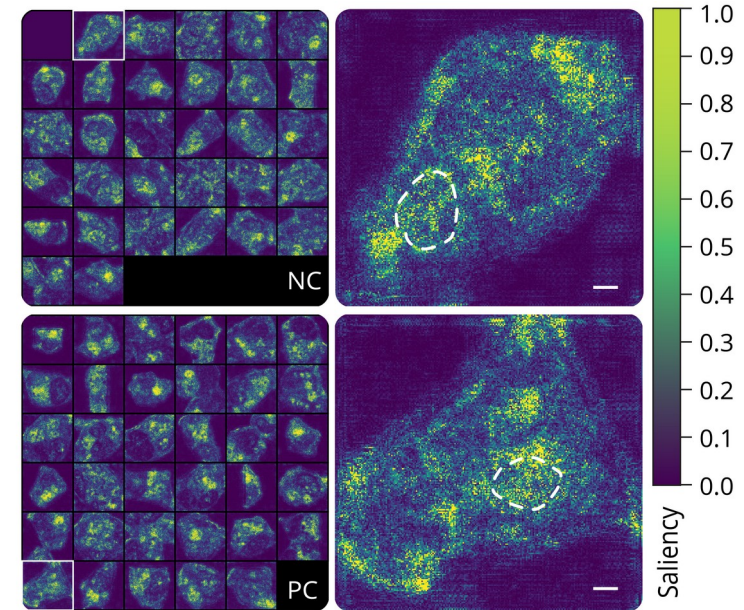
CORE · 4 OF 6

Classify: image model

ALPHA



Input: single-cell crops from negative (NC) and positive (PC) control wells



Saliency: where MaxViT looked: at the vacuole in PC cells

SUB-TOOLS INSIDE THIS MODULE

Activation Maps

Show which pixels drove each decision

Feature Explorer

Which measurements separate the classes best

Classifier Evaluation

Accuracy and mistakes per class and per plate, on unseen wells

Training Runs

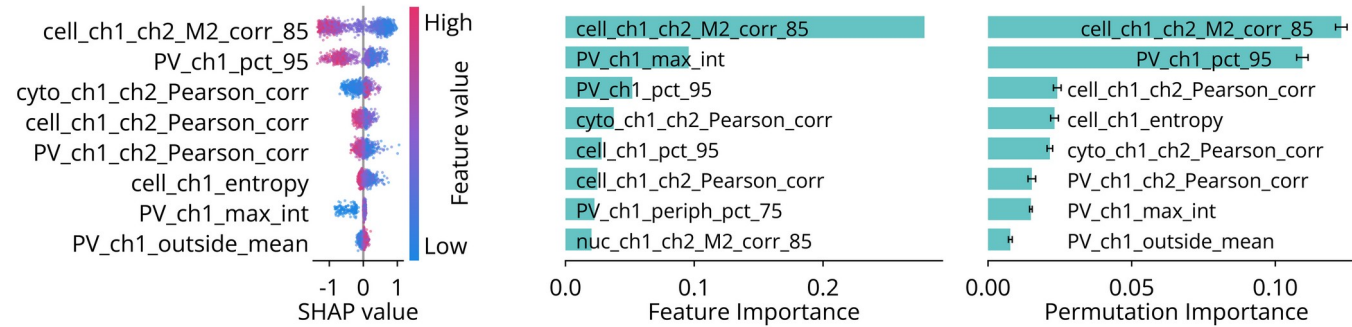
Compare training curves between runs

Explain CV Model

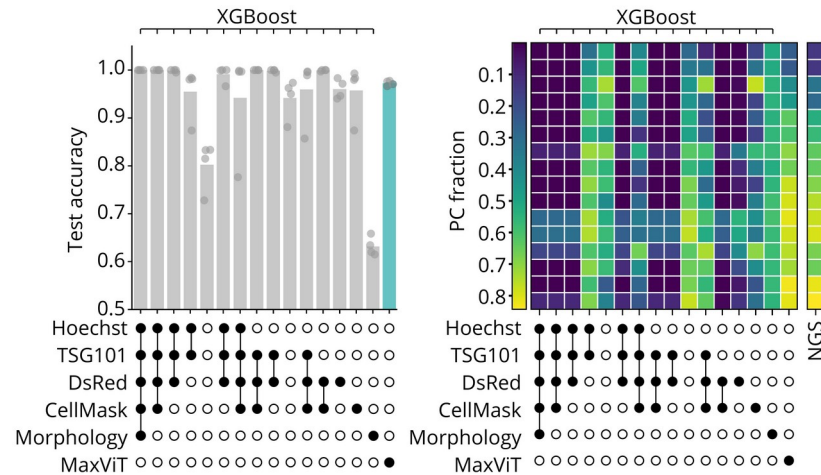
Explain an image model in terms of measurements



Classify: measurement model



XGBoost on the measurements: which measurements decide the class, three ways of ranking them



Accuracy with different channel combinations (grey, XGBoost) vs the image model (teal, MaxViT)



CORE · 5 OF 6

STABLE

Map Barcodes: which guides, which well

Reads the sequencing data and works out which gRNAs are in each well, and how much of each.

WHAT YOU CAN DO

- Reads the row, column and gRNA barcodes from the sequencing files
- You give it the barcode lists and where they sit in the read
- Paired-end or single reads; set how many mismatches to allow
- Quick test on a small part of the file first

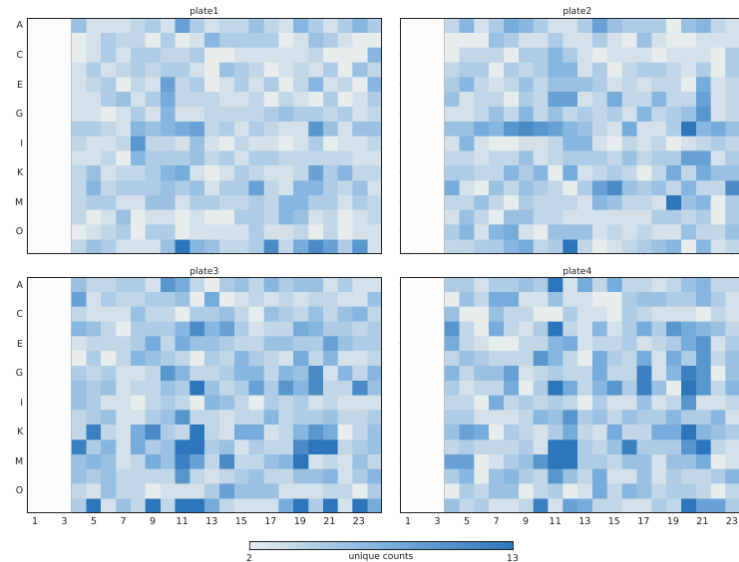
OUTPUT

- Guide counts per well
- QC table and run manifest
- gRNA fraction per well → input to Regression

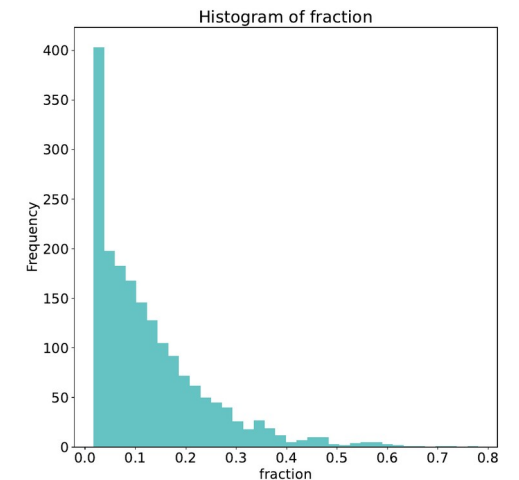
SUB-TOOLS INSIDE THIS MODULE

Barcode QC

Reads per well, reads that did not match, library coverage: and the cut-off that gives the expected number of gRNAs per well



Unique gRNAs per well, four plates



gRNA fraction per well



Regression: from wells to genes

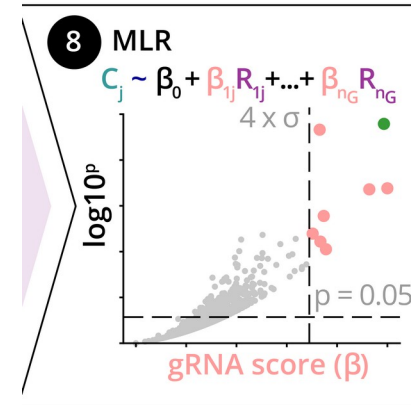
Works out which genes change the phenotype, and ranks them.

WHAT YOU CAN DO

- Pairs each well's score with the fraction of each gRNA in that well
- Default: a mixed model where the gRNAs of a gene are grouped
- Other models for counts, fractions or skewed data
- Permutation test: shuffles gRNAs to check a hit is not chance

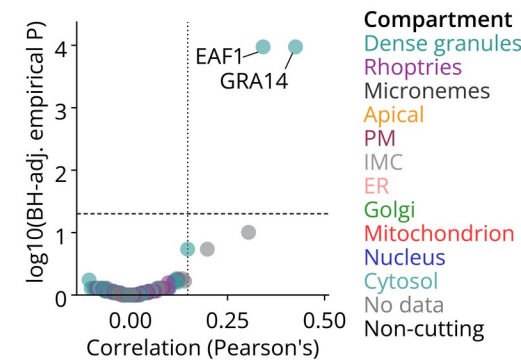
OUTPUT

- A ranked table of genes with effect size and p-value
- Volcano, top effects and quality-check plots



G: gRNA
R: gRNA read f
C: Classifier score
 β : gRNA score
 β_0 : Intercept
W: Well
i: gRNA index
j: Well index

The model: a well's score = its gRNA fractions × each gene's effect (preprint Fig. 2)



The result: EAF1 and GRA14 stand out (preprint Fig. S7)



Regression: choose the model

21 model families, each explained in plain words, with what it assumes about your data.

WHAT YOU CAN DO

Mixed: the gRNAs of a gene grouped, genes allowed to vary

OLS / WLS: classic least squares, optionally weighted by cells per well

GLM, Poisson, beta, logit: for counts, fractions and yes/no readouts

Lasso, ridge, elastic net, group lasso, horseshoe: when there are many more genes than wells

Robust, quantile, RRA: when a few wells are extreme

OUTPUT

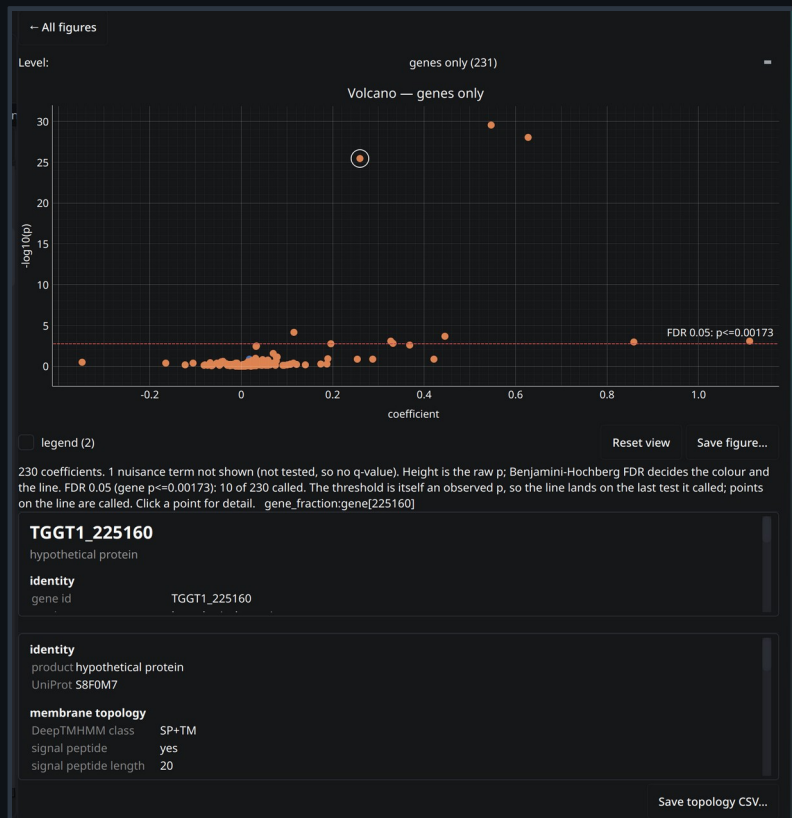
- Plate, row and column covariates; permutation test on gRNAs
- The formula and what it assumes, shown for the chosen model

```
ols — parametric: least squares: normal errors of constant variance, every well equal  
auto — chosen from the response by check_distribution  
mixed — parametric: guides nested in genes as random effects, those effects normal; guides that disagree widen their gene's interval  
beta — parametric: a fraction strictly inside 0 and 1, beta distributed; exact zeros and ones have to be moved off the boundary first  
elasticnet — parametric: both penalties mixed by l1_ratio: sparse, but correlated guides of a gene are kept or dropped together; no p value from the fit -- ranked by bootstrap selection frequency  
glm — parametric: a family chosen from the response, whose mean-variance relationship is then assumed to hold  
group_lasso — parametric: a gene's guides penalised as one block, so a gene is in or out as a whole rather than by its best guide; no p value from the fit -- ranked by bootstrap selection frequency  
hinge — parametric: a linear boundary between a positive and a negative class: the response is thresholded, so the question is which class a well is in, not by how much  
horseshoe — parametric: Bayesian shrinkage assuming most guides do nothing and a few do a lot; the interval is a posterior, not a p-value  
lasso — parametric: normal errors plus an L1 penalty: the truth is sparse and one guide of a correlated group carries the effect; no p value from the fit -- ranked by bootstrap selection frequency  
logit — parametric: a fraction as a binomial proportion, log-odds linear in the guides, weighted by the cells behind the well  
ols — parametric: least squares: normal errors of constant variance, every well equal  
poisson — parametric: counts, variance equal to the mean, log cell count as exposure so the effect is on a per-cell rate  
probit — parametric: the same binomial proportion as logit through a normal link, so the tails are lighter  
quasi_binomial — parametric: a fraction with binomial mean-variance, freely rescaled, so overdispersion is absorbed not modelled  
ridge — parametric: normal errors plus an L2 penalty: the effect is shared across correlated guides rather than assigned to one  
wls — parametric: least squares weighted by cell count: normal errors whose variance falls as a well holds more cells  
huber — robust/semiparametric: squared loss near the centre and linear in the tails, so a handful of bright wells cannot set a coefficient  
quantile — robust/semiparametric: a quantile of the response rather than its mean, which assumes nothing about the shape of the errors; tau=0.9 asks whether the guide moves the TOP of the distribution  
rlm — robust/semiparametric: a linear model fitted by an M-estimator, so extreme wells are downweighted rather than assumed absent  
spline — robust/semiparametric: least squares, but each continuous covariate is free to bend; guides keep one coefficient each, so the volcano is unchanged. Semiparametric, not fully nonparametric  
rra — rank-based: only the ORDER of the wells, so no distribution at all; its p value is a permutation p value over guide ranks
```

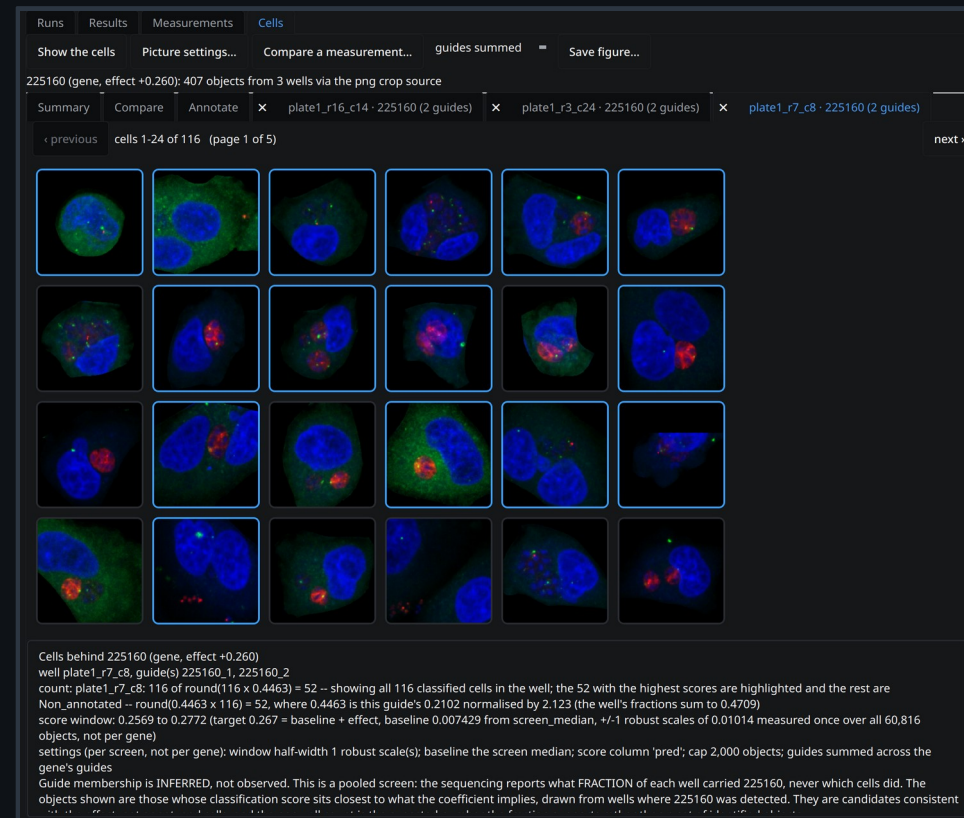
Regression type: every model with the assumption it makes (OLS selected)



Regression: pick a gene, see its cells



Interactive volcano: click a point. EAF1 (TGGT1_225160) is selected, its gene card below



Cells tab: the cells in an EAF1 well whose scores fit the effect, with why they were picked

SUB-TOOLS INSIDE THIS MODULE

Volcano Explorer

Publication version of the volcano;
vector export

Investigate Hit

Test the hit at the cell level against
control wells

Hit List

Ranked genes with intervals and gRNA
agreement

Prediction Profiler

How the prediction changes as one
gene's fraction changes



Regression: which assumptions hold?

Every run checks its own assumptions and says, in red, which ones fail and what to do.

WHAT YOU CAN DO

Equal variance: Breusch-Pagan test on the residuals

Normal residuals: skew, kurtosis, D'Agostino test

Independence: Durbin-Watson; wells on one plate share handling

Influence: leverage and Cook's distance per well

Collinearity: VIF and condition number

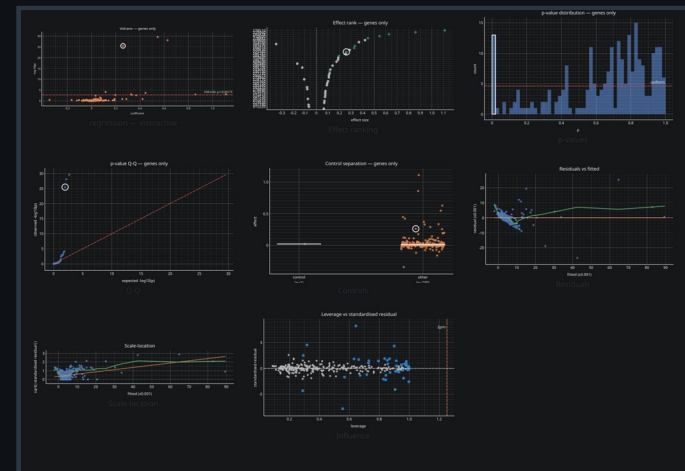
Calibration: p-value histogram and genomic inflation

OUTPUT

- A run summary: the answer, what was fitted, each assumption with its verdict
- Residual, Q-Q, scale-location and influence plots

ASSUMPTIONS, EACH WITH ITS TEST AND ITS VERDICT	
equal variance	Breusch-Pagan $p = 0.000941$ (REJECTED at 0.05); White's test was not run: it squares every column pair, so it is attempted only on designs of at most 30 columns and skipped when singular — at least one test rejects equal variance, so the standard errors here are the wrong width; cov_type='HC3', or weighting wells by cell count, is the fix
normality of residuals	skew -0.10, excess kurtosis +27.41 over 425 residuals; D'Agostino $K^2 p = 1.22e-30$ (REJECTED at 0.05)
independence / clustering	Durbin-Watson = 1.84 (2 is none) — no first-order autocorrelation, wells within a plate share reagents, imaging and handling; this fixed-effects fit ASSUMES independent errors and models position with rowID / columnID terms rather than a plate random effect — regression_type='mixed' is the fit that relaxes it
influence and leverage	max leverage 1 against the 2p/n guide of 1.25 (0 above it); max Cook's D 8.6, 42 observation(s) above the 4/n rule of 0.00941
multicollinearity	max VIF 33.9, 15 above 10; scaled condition number 46.59 — moderate to strong collinearity — standard errors are inflated; 0 predictor pair(s) correlated at 1; 1 at 0.95 (worst 0.35)
THE CALL	
correction	fdr_bh (Benjamini-Hochberg FDR)
alpha	0.05 (fdr_alpha, the level the correction targets)
coefficients tested	230 of 231 rows are hypotheses; the other 1 are covariates (the intercept and the row / column terms), which are fitted so the real effects are estimated cleanly and are not themselves tested
coefficients called	10 — q value below 0.05
critical raw P	0.001726 — every test with a raw P at or below this was called by fdr_bh at alpha=0.05. It is NOT alpha, and it is far below it
genomic inflation	0.4601 — DEFLATED: the tests are conservative; the null is not flat and the P values are larger than the evidence warrants
effect-size cut	none — 1 control coefficient(s) is not enough to measure a spread
not applicable here	2 field(s) — positive control rank, positive control percentile. Why each does not apply is under NOT APPLICABLE, AND WHY at the end.

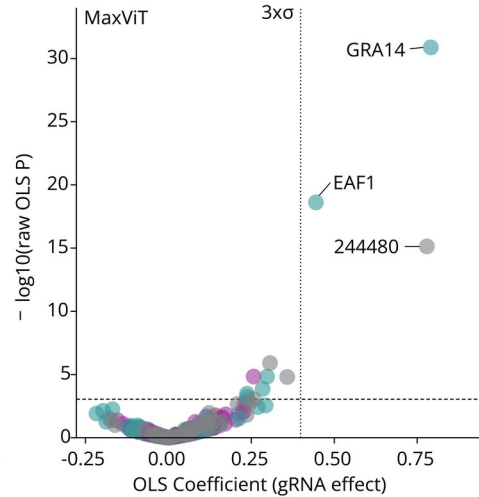
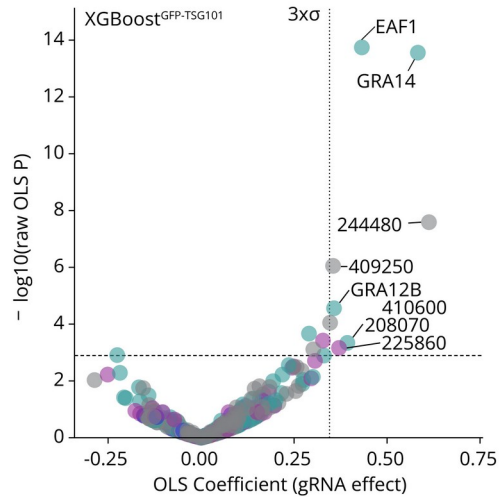
Summary tab: each assumption with its test and verdict; equal variance and normality rejected (red)



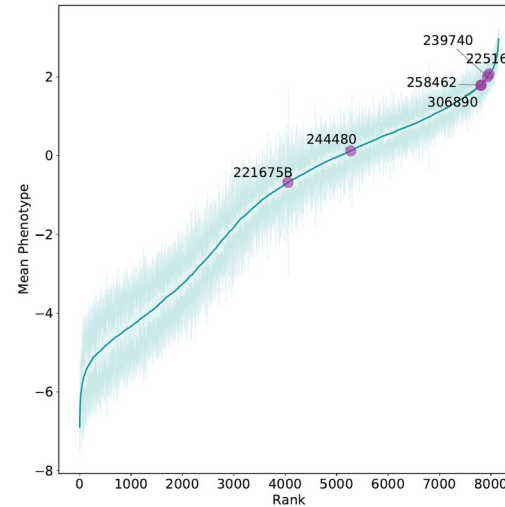
The diagnostic plots behind the verdicts



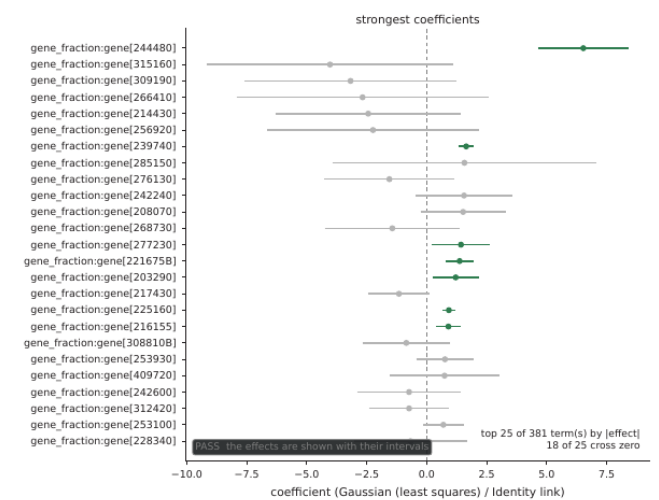
Regression: the hit list



Volcanos from the screen: XGBoost (left) and MaxViT (right) scores both find EAF1 and GRA14 (preprint Fig. 5)



Genes ranked by phenotype



Strongest effects with 95% intervals

SUB-TOOLS INSIDE THIS MODULE

Volcano Explorer

Click a point to see the gene; export for publication

Hit List

Ranked genes with confidence intervals and gRNA agreement; export as a table

Investigate Hit

Jump from a gene to its wells and cells

Diagnostics

Checks that the model fits the data (plate, row and column effects)

Methods & Results

Drafts the methods text from the run itself

Prediction Profiler

How a prediction changes across one input

Parameter Sweep

Re-run across settings to see how stable the hits are

02

Data

Bring data in, plan experiments, and check that a run can be trusted.



Import



Embeddings



Run Compare



Experiment Design



Power / Design



Dose-Response



QC



Import: bring outside data in

Turns images, masks and tables from other tools into a spaCR project.

WHAT YOU CAN DO

- Pair images, instance masks and a measurement CSV
- Match their column names to spaCR's, converting units
- Choose what happens when names collide
- Preview before anything is written; save and reuse the mapping

OUTPUT

- A spaCR project that records where the data came from

Their column	spaCR column	Transform	Unit in	Unit out	Note
AreaShape_Area_um2	foreign_areashape_area_um2	area	um^2	px^2	unit um^2 read from the column name; spaCR measures in px^2, so this needs a pixel size
MeanIntensity_ER	foreign_meanintensity_er	identity			copied unchanged; declare unit_in/unit_out if it needs scaling
Metadata_Well	foreign_metadata_well	identity			copied unchanged; declare unit_in/unit_out if it needs scaling
cell_area	foreign_cell_area	identity			name suggests a area, but no unit is declared — set unit_in (and transform=area) if these are not already in px^2

Column mapping: their name → spaCR column, with unit conversion

Source	Target	Plate	Well	Field	Channel	Z	T	From well	From field	From channel	Z handling	Status
fov01_C1.tif	plate1_A01_T0001F001L01A01Z01C01.tif	plate1	A01	1	1	1	1	A01	images/fov01	C1	keep	planned
fov01_C2.tif	plate1_A01_T0001F001L01A01Z01C02.tif	plate1	A01	1	2	1	1	A01	images/fov01	C2	keep	planned
fov01_C3.tif	plate1_A01_T0001F001L01A01Z01C03.tif	plate1	A01	1	3	1	1	A01	images/fov01	C3	keep	planned
fov01_C4.tif	plate1_A01_T0001F001L01A01Z01C04.tif	plate1	A01	1	4	1	1	A01	images/fov01	C4	keep	planned
fov02_C1.tif	plate1_A01_T0001F002L01A01Z01C01.tif	plate1	A01	2	1	1	1	A01	images/fov02	C1	keep	planned
fov02_C2.tif	plate1_A01_T0001F002L01A01Z01C02.tif	plate1	A01	2	2	1	1	A01	images/fov02	C2	keep	planned
fov02_C3.tif	plate1_A01_T0001F002L01A01Z01C03.tif	plate1	A01	2	3	1	1	A01	images/fov02	C3	keep	planned
fov02_C4.tif	plate1_A01_T0001F002L01A01Z01C04.tif	plate1	A01	2	4	1	1	A01	images/fov02	C4	keep	planned

Format Converter: plan for turning vendor files into spaCR's TIFF layout

SUB-TOOLS INSIDE THIS MODULE

Import Images

Reads any microscope's folder and works out the naming itself

Format Converter

Turns Nikon, Zeiss and Leica files into spaCR's layout

External Masks

Measure masks made elsewhere, without re-segmenting



Embeddings: a fingerprint per cell

Gives every cell a label-free feature vector from a pretrained image model, next to the measured features.

WHAT YOU CAN DO

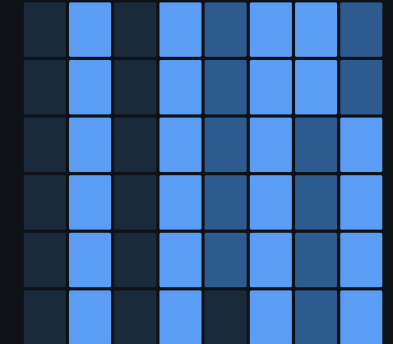
- Crops from the project (filter by object class or plate) or from a folder
- Choose the network: ResNet-18 (default), ResNet-50, ConvNeXt, ViT
- Each stain is described separately, or all together
- Runs on the graphics card when there is one



cell crops



**pretrained
backbone**
e.g. ResNet-18



one row of numbers per cell

OUTPUT

- One row of numbers per cell, next to its measurements
- Which network was used is recorded

Use it for clustering and UMAPs without choosing features, finding cells that look like a hit, and checking whether a classifier is picking up something the measurements miss.



Run Compare: what changed?

Puts two runs of the same project side by side, so you know why the results moved.

WHAT YOU CAN DO

- Pick a project and any two of its runs
- Settings diff, grouped by settings panel
- Object, well and field counts per plate
- Hits that appeared, vanished or changed rank
- Incompatible runs are blocked unless you choose 'compare anyway'

OUTPUT

- Side-by-side comparison tables

Settings

pathogen_model	cpsam → toxoplasma_pv_v3
cell_diameter	90 → 110
regression_type	ols → mixed

Counts per plate

cells	182,410 → 176,902
pathogens	40,112 → 52,774
wells	384 → 384

Hits

appeared	3 genes
vanished	1 gene
rank changed	12 genes

Illustration of the three comparison panels (values made up)



Experiment Design: plan the plate

Lay out conditions, controls and replicates before you image anything.

WHAT YOU CAN DO

- Assign conditions and control roles to wells
- Warnings for capacity, too few replicates, and row / column / edge confounding
- Randomise with a seed, so the layout can be rebuilt exactly

OUTPUT

- A plate map that later spaCR steps use directly
- The settings used, so the layout can be rebuilt

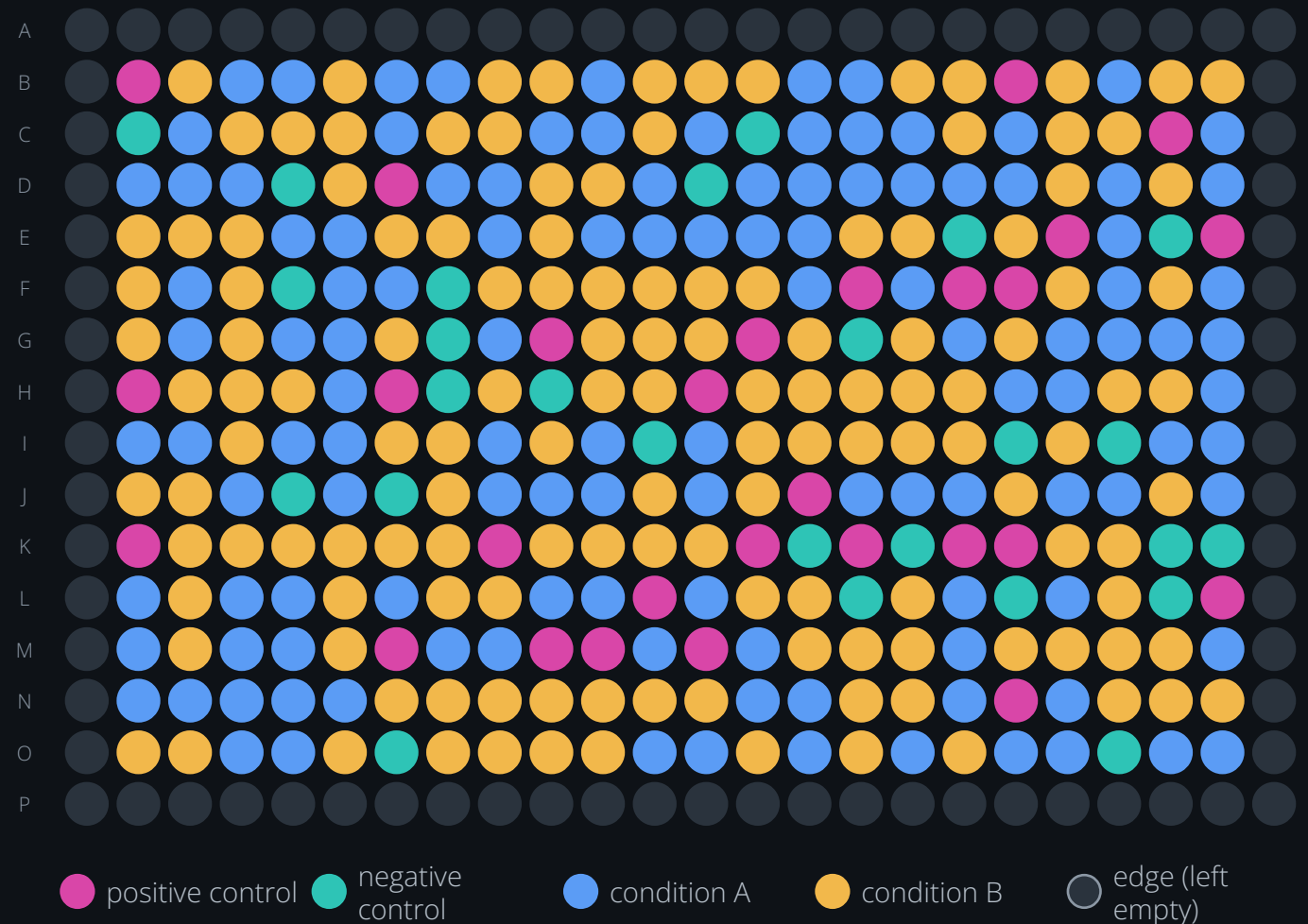


Illustration: a randomised 384-well layout with edge wells left empty



Power / Design: how big a screen?

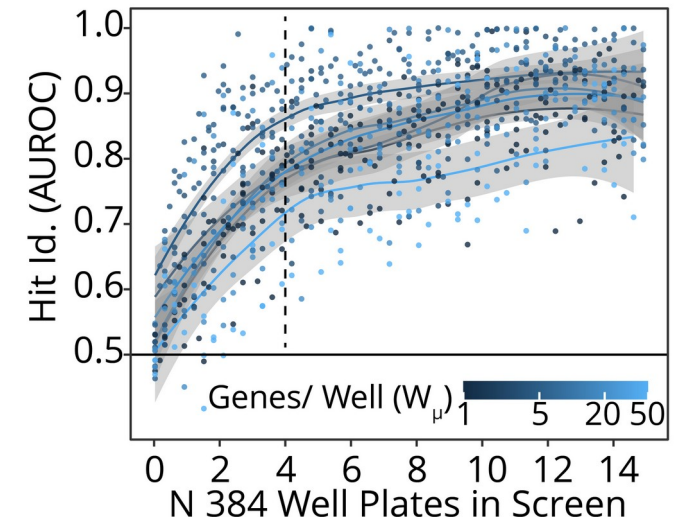
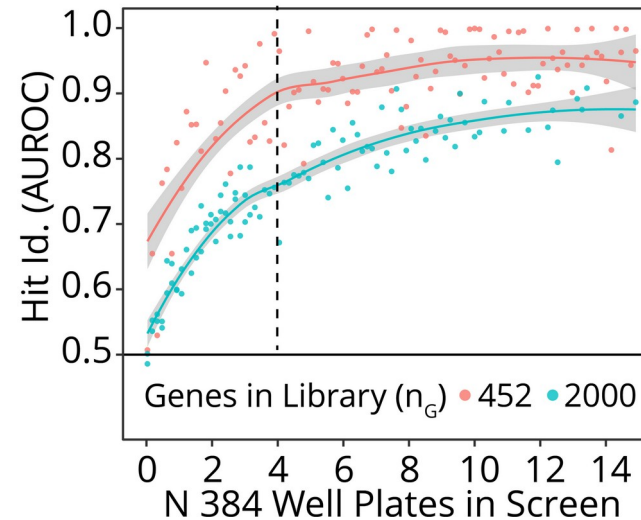
Estimates how many cells per well and how many plates you need to find a real hit.

WHAT YOU CAN DO

- Enter the library: genes, gRNAs per gene, constructs per well
- Enter the plate format and number of plates
- Enter the classifier's background rate and the effect you expect
- Simulates the screen many times to see how often the real hit is found

OUTPUT

- Detection probability vs cells per well
- A one-sentence answer, with caveats



spaCRPower simulations: hit detection vs plates, by library size and genes per well (preprint Fig. 1D-E)



Dose-Response: honest EC50s

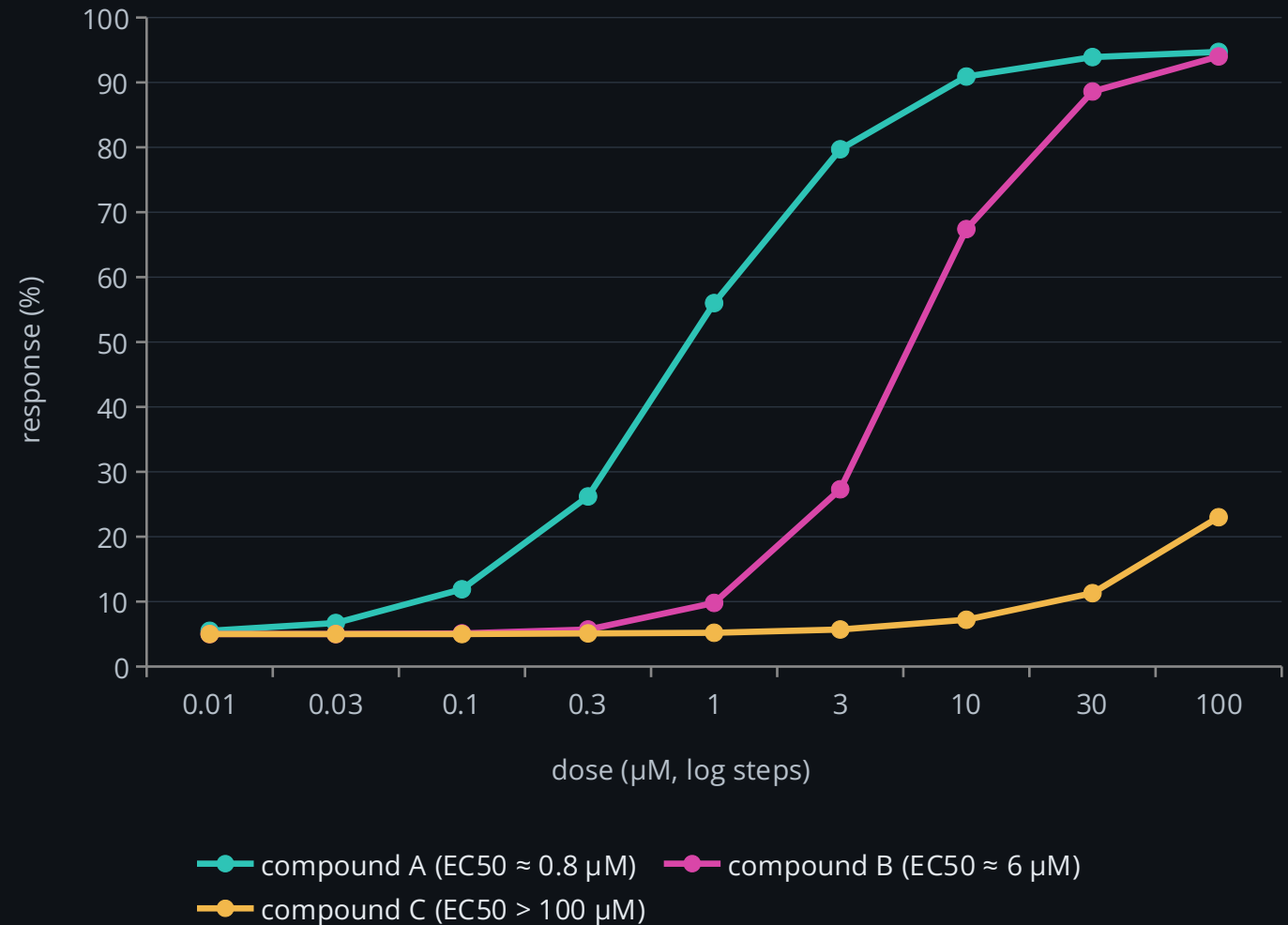
Fits dose-response curves per gene or compound, and says so when an EC50 can't be pinned down.

WHAT YOU CAN DO

- Choose concentration, response and grouping columns
- Fits the standard S-shaped curve
- A confidence interval on every EC50

OUTPUT

- Curves on a log-dose axis
- EC50 table with CI. Out-of-range → a bound ('EC50 > 30 μ M'); failed fits listed as 'refused'
- Warns when a curve does not fit well



Schematic of the output (simulated curves)



DATA · 7 OF 7

ALPHA

QC: is this run usable?

Five stored checks on one page. Reads what the run already wrote; recomputes nothing.

WHAT YOU CAN DO

- Segmentation QC scorecards
- Pixel size recorded
- Classifier tested on truly unseen wells
- Plate effects (row / column / edge)
- Annotator agreement

OUTPUT

- Verdict cards: ok · warn · fail · missing · stale
- Nothing is written to disk

Segmentation

OK

98% of fields pass

Units

OK

0.325 μm / pixel

Classifier test

WARN

same well used to train and test

Plate effects

FAIL

row gradient, $p = 0.41$

Annotator agreement

MISSING

no second annotator

Illustration of the dashboard's verdict cards (values made up)

SUB-TOOLS INSIDE THIS MODULE

Layer Viewer

Images and masks as layers you can switch on and off

Control Charts

Follow a control over time and flag drift

Outliers

Find unusual cells or wells

03

Tools

Point these at a project: fix masks, stitch tiles, explore cells, gate populations, plot.



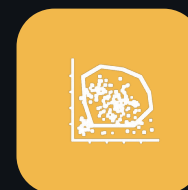
Make Masks



Align & Stitch



Image UMAP



Gate Editor



Graph Builder



TOOLS · 1 OF 5

BETA

Make Masks: fix masks by hand

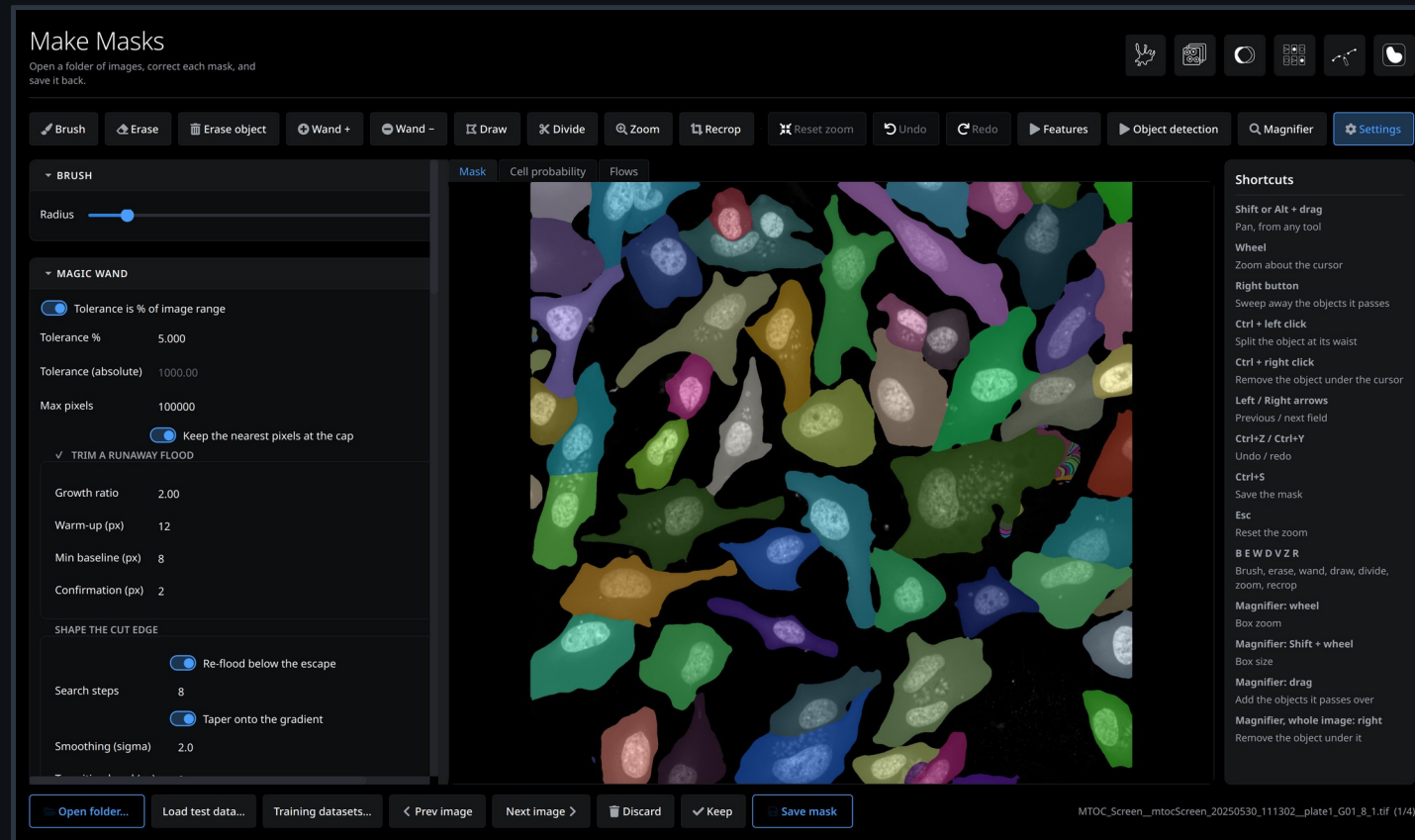
Correct segmentation by hand, and build training data for your own Cellpose model.

WHAT YOU CAN DO

- Nine tools: brush, erase, erase object, magic wand + / -, draw, divide, zoom, recrop
- Live magnifier: a box that zooms in and outlines cells as you move
- Object detection on the whole field (Cellpose or a simple threshold)
- Invert the image so dark objects can be detected too
- Features: measure the masks you drew, in one step

OUTPUT

- Curated masks, ready for training or measuring
- A log of every edit, so curated masks are told apart from model output



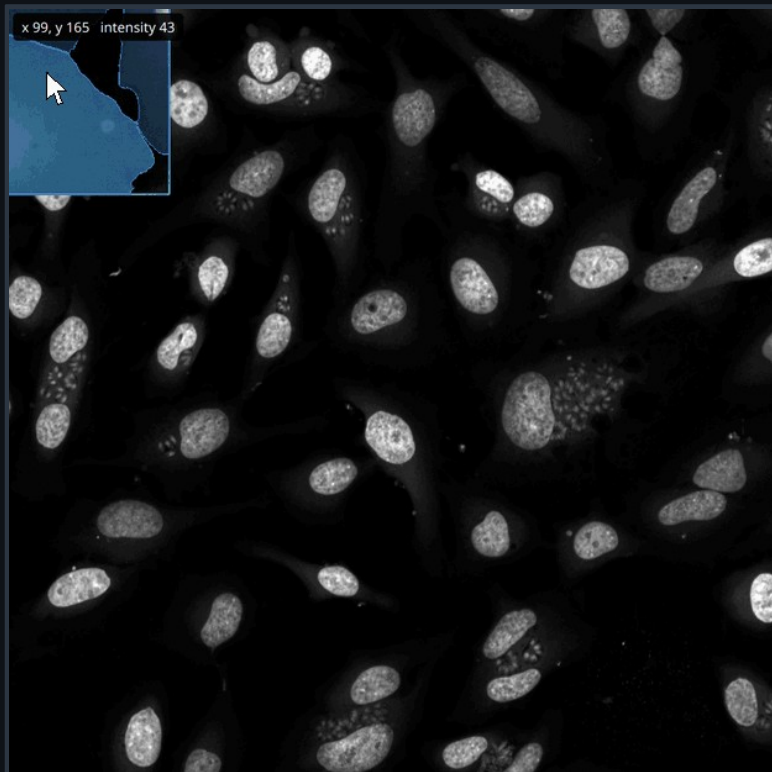
The current Make Masks: tools across the top, settings left, shortcuts right



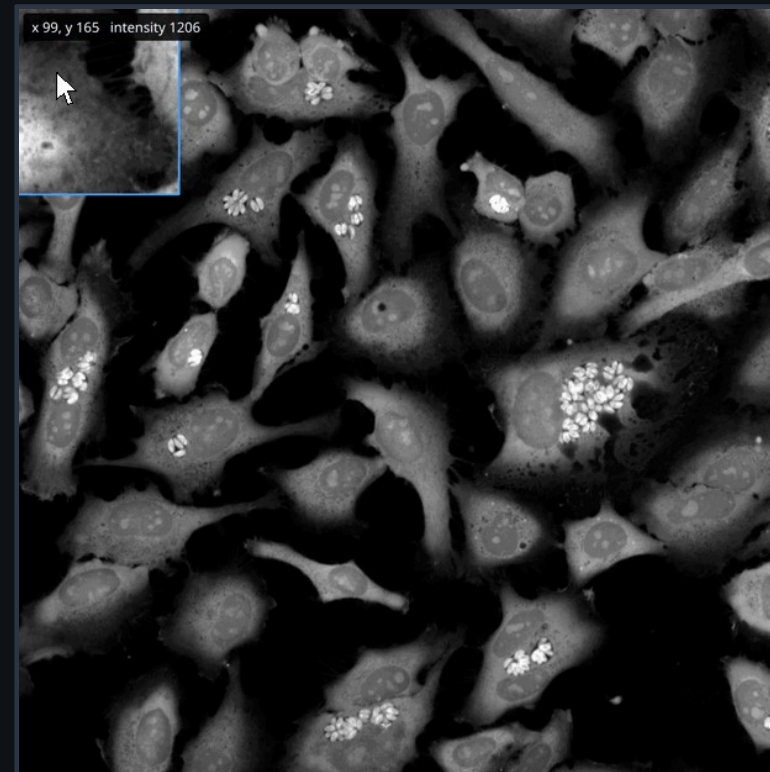
TOOLS · 1 OF 5

BETA

Make Masks: live magnifier



Cell from Hoechst: whole cells found from the nuclear (Hoechst) image



Toxoplasma from Cell Mask: vacuoles found from the cell-mask image

Box

follows the mouse, zoomed and contrast-adjusted

Model

Cellpose-SAM, your own model, or a threshold, outlining live

Click

adds what the box found; every click can be undone

Drag

along a cell bigger than the box joins its pieces into one object



TOOLS · 1 OF 5

BETA

Make Masks: Features

Measure the masks you drew, without leaving Make Masks.

WHAT YOU CAN DO

- One row per field; columns are the image channels and the mask types
- Fill it by dropping files, clicking a cell to pick a file, or one filename pattern
- Press Measure: the run goes through the Measure module itself

OUTPUT

- The same measurements database and crops a Measure run makes, from your hand-corrected masks

The files

Drop images and masks here, or write one regex that sorts them. Rows are fields; columns are channels and mask types.

Plate name

drawn|

Channels

1

☒ Cell mask

☐ Nucleus mask

☐ Pathogen mask

☐ Organelle 1

Regex

(?P<field>-+?(?P<plate>plate\d+)(?P<well>[A-P]\d{2})\d+)(?P<channel>\d+)\.tif

Re-apply to all files

4 file(s) assigned, 0 left over.

	Field	Well	Field #	Channel 1	Cell mask
1	MTOC_Screen__mtocScreen_20250530_111302__plate1_G01_8	G01	1	MTOC_Screen__mtocScreen_20250530_111302__plate1_G01_8_1.tif	MTOC_Screen__mtocScreen_20250530_111302__plate1_G01_8_1.tif
2	MTOC_Screen__mtocScreen_20250531_145355__plate2_B04_18	B04	2	MTOC_Screen__mtocScreen_20250531_145355__plate2_B04_18_1.tif	MTOC_Screen__mtocScreen_20250531_145355__plate2_B04_18_1.tif
3	MTOC_Screen__mtocScreen_20250601_172718__plate3_H06_4	H06	3	MTOC_Screen__mtocScreen_20250601_172718__plate3_H06_4_1.tif	MTOC_Screen__mtocScreen_20250601_172718__plate3_H06_4_1.tif
4	MTOC_Screen__mtocScreen_20250601_172718__plate3_I10_20	I10	4	MTOC_Screen__mtocScreen_20250601_172718__plate3_I10_20_1.tif	MTOC_Screen__mtocScreen_20250601_172718__plate3_I10_20_1.tif

Add field

Remove selected

Clear

Nothing was left over.

Input_Experiment

Mask_Channel Mapping

Illumination Correction

4 field(s), 1 channel(s), 1 mask type(s). The settings the table decides are filled in below.

What the run writes appears here.

Measure

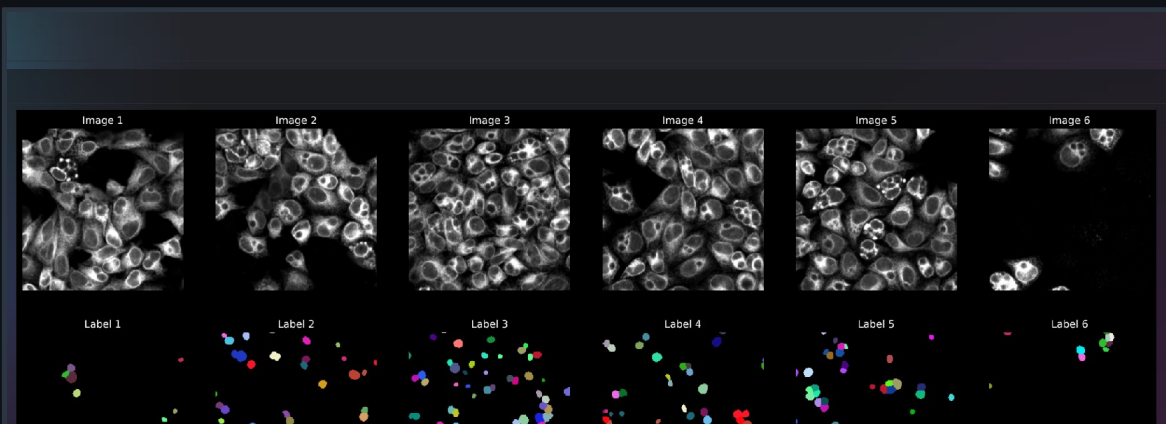
Features: four fields with their cell masks, ready to measure



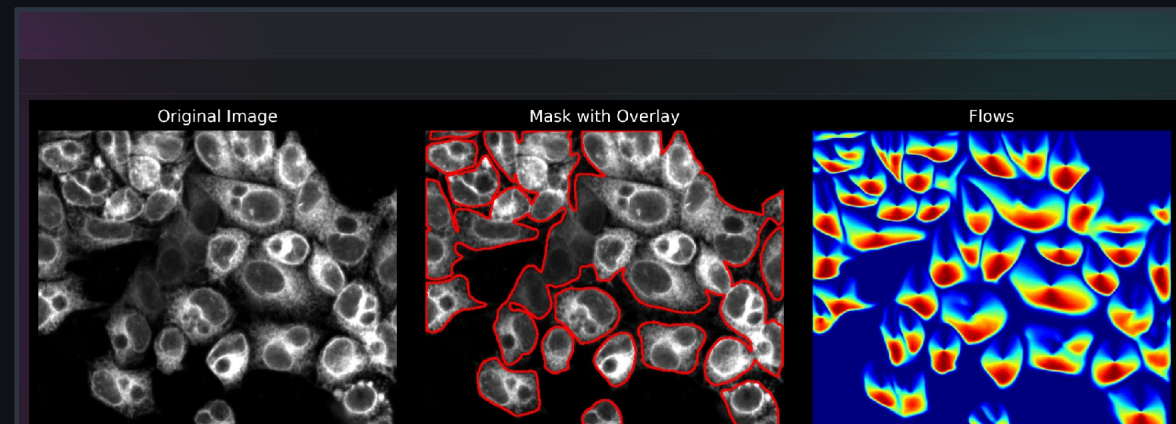
TOOLS · 1 OF 5

BETA

Make Masks: train and compare models



Cellpose Workbench, training: image / label pairs it will learn from



Cellpose Workbench, applying a model: image, mask overlay and flows

SUB-TOOLS INSIDE THIS MODULE

Cellpose Workbench

Fine-tune Cellpose on your masks, then mask the whole folder

Model Compare

Two models' masks side by side, and how much they agree

Model Zoo

Browse, download and test trained models

Curate

Edit masks and tracks, with an edit log

Napari Bridge

Edit a mask in napari and bring it back



TOOLS · 2 OF 5

STABLE

Align & Stitch: tiles to mosaic

Stitches overlapping tiles into one large image without running out of memory.

WHAT YOU CAN DO

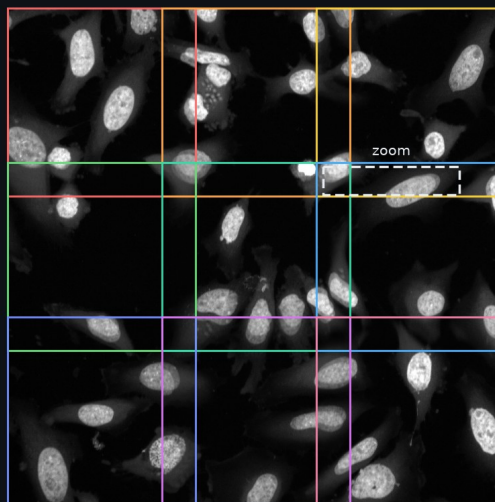
- Set grid, overlap, tile order and reference channel
- Plans first, reading only the overlaps: fast on huge plates
- Finds how far each tile is off by matching nuclei in the overlaps
- Shows how sure it is about every tile
- Tells you the final size and memory needed before it starts

OUTPUT

- The stitched mosaic, written piece by piece so memory stays low
- Where every tile ended up, saved with the project

1 Tiles as acquired

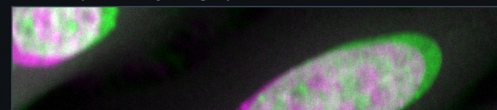
Placed where the stage said: each tile is up to 20 px off, so nuclei in the overlaps do not line up.



2 Overlap matched

Zoom on the strip two neighbouring tiles share. Tile A magenta, tile B green: white where they agree.

Before: placed by stage position



tile B moved 19 px to match tile A

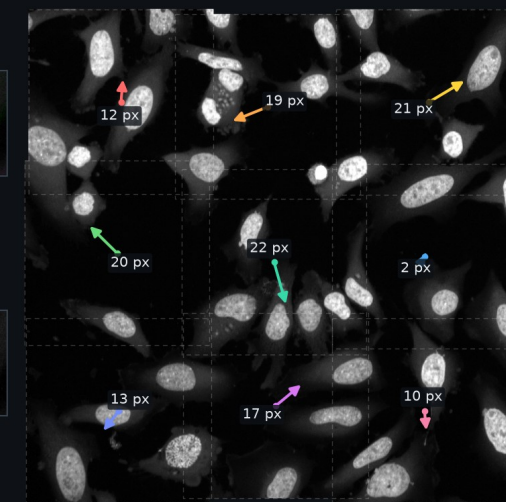
After: shifted until the nuclei line up



■ tile A ■ tile B ■ both

3 Stitched mosaic

Every tile moved to where its nuclei match. One seamless image; arrows show each move.



arrows drawn 8x longer · all 9 tiles placed, 20/20 overlaps matched

Real Hoechst image, cut into 9 overlapping tiles with simulated stage error; shifts found and blended by spaCR (spacr.align).

A real Hoechst image cut into 9 overlapping tiles, each up to 20 px off (simulated); spaCR put every tile back where it came from



TOOLS · 3 OF 5

BETA

Image UMAP: map all cells

Places every cell on a 2-D map where similar cells sit together: with the real images.

WHAT YOU CAN DO

- UMAP, t-SNE or PCA maps
- Finds groups of similar cells automatically
- Embed by controls (positive vs negative groups)
- Can choose the map settings for you
- Click a point to see that cell's crop

OUTPUT

- Scatter coloured by cluster or condition, with thumbnails
- Table with a cluster column per cell



5,000 real cells from TSG101 screen plate 1: they fall into groups by number of nuclei and vacuoles

SUB-TOOLS INSIDE THIS MODULE

Image Scatter

Any two measured features, with the crop shown on hover

PCA

Which measurements drive the main differences



TOOLS · 4 OF 5

ALPHA

Gate Editor: gate cells like FACS

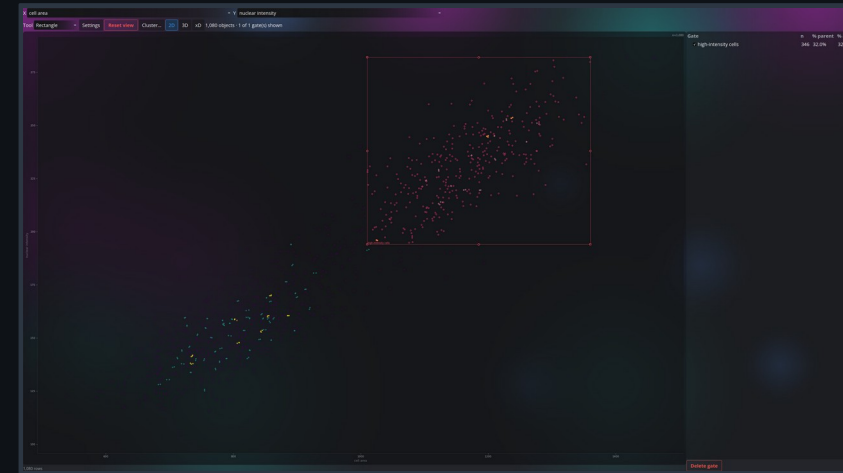
Select cell populations the way a flow cytometrist gates: in 1, 2 or 3 dimensions.

WHAT YOU CAN DO

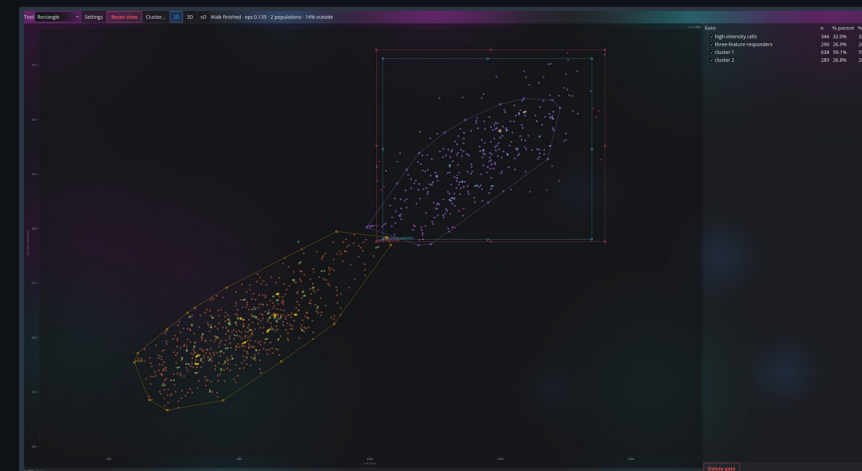
- Load tables from one or several projects, or a CSV
- Threshold, rectangle, oval, polygon, wand gates in 1-D and 2-D
- Box, cylinder and prism gates in 3-D
- Nest and combine gates; count and % of parent for each
- Publish a gate as a filter every linked view follows

OUTPUT

- A gating strategy you can reuse on other plates
- Gate-membership columns in the project; PNG / PDF plots



2-D rectangle gate on cell area vs nuclear intensity



Cluster walk: populations found and gated automatically



TOOLS · 5 OF 5

BETA

Graph Builder: plots without code

Build a plot by dragging columns onto axes. The chart type follows the data.

WHAT YOU CAN DO

- Six drop zones: x, y, colour, size, facet row, facet column
- Histogram, bar, scatter, box or heatmap chosen for you; violin and line on request
- Brush points to select cells: linked to the UMAP and plate map
- Local filters without touching the project

OUTPUT

- Interactive charts, exportable as figures

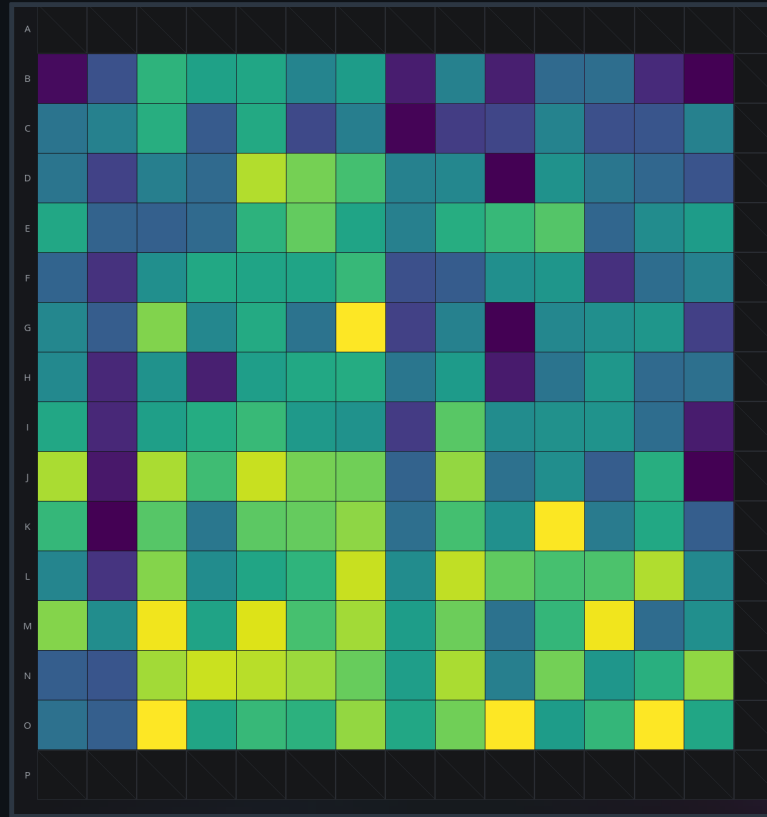
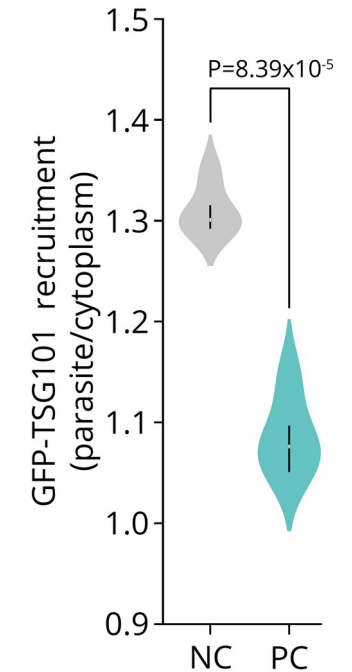


Plate Viewer: mean cell intensity per well



Violin plot of per-well values

SUB-TOOLS INSIDE THIS MODULE

Plate Viewer

Plate heatmaps and edge-effect detection

Small Multiples

One plot per group, shared or independent axes

04

Assays

Ready-made readouts for common parasite assays.



Plaque



Recruitment



Invasion



Replication



Plaque Assay: count and size plaques

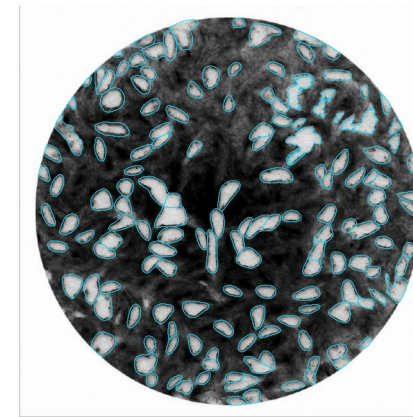
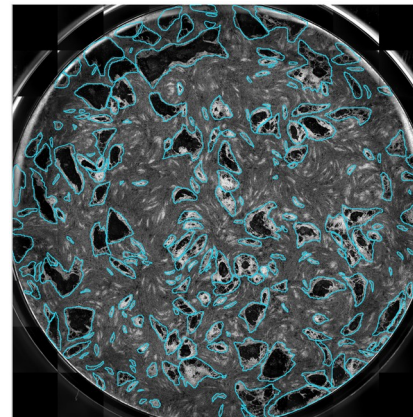
Finds every plaque in whole-well images and measures it in mm².

WHAT YOU CAN DO

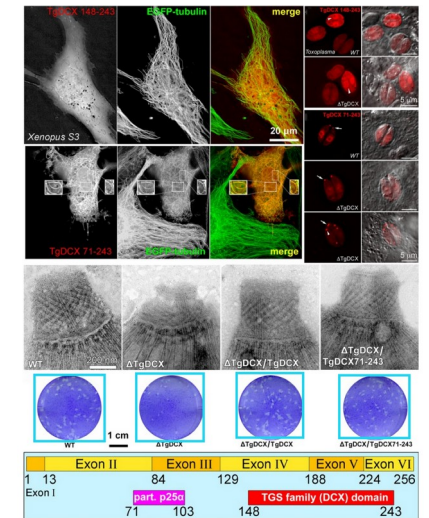
- Segments plaques with the Model Zoo plaque model (Cellpose-SAM)
- A YOLO detector finds and splits wells first
- Areas converted to mm² from the well diameter
- 'Figure' mode for plaque images taken from papers
- Can reuse existing masks

OUTPUT

- A table of every plaque, and a summary per well
- Area, perimeter, diameter, eccentricity, solidity per plaque



Curated plaque outlines: our microscope (left) and a published figure (right)



Wells found in a figure



Recruitment: marker at the vacuole

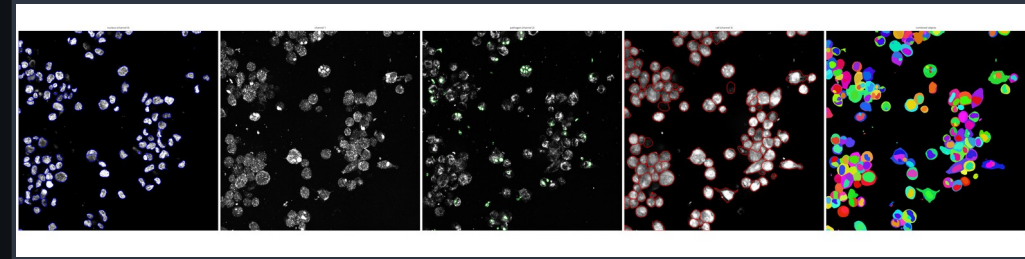
How much does a fluorescent marker gather at the parasite vacuole compared with the host cytoplasm?

WHAT YOU CAN DO

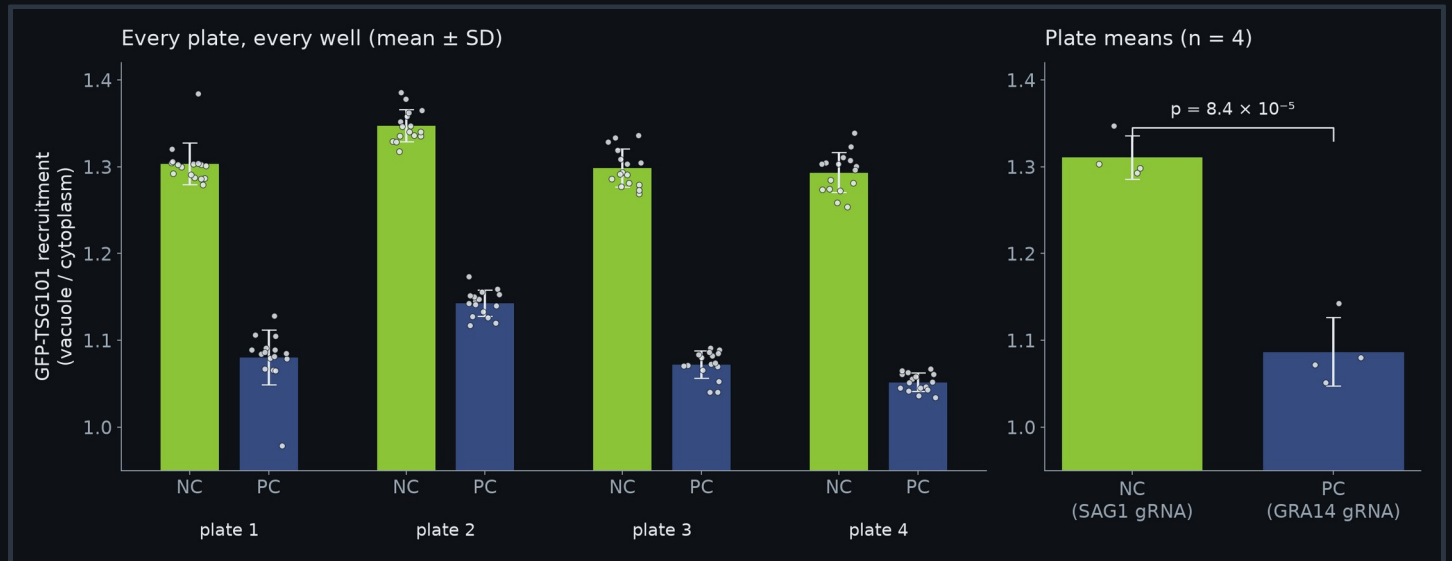
- Map wells to cell type, pathogen and treatment
- Size and intensity filters per object class
- Choose the channel of interest
- Minimum number of cells per well

OUTPUT

- Tables per cell and per well
- Bar plots of recruitment by condition, with per-well points



One field: channels and the cell / nucleus / parasite masks



GFP-TSG101 recruitment to the vacuole, NC vs PC, 4 plates (preprint data)



Invasion: attached or invaded?

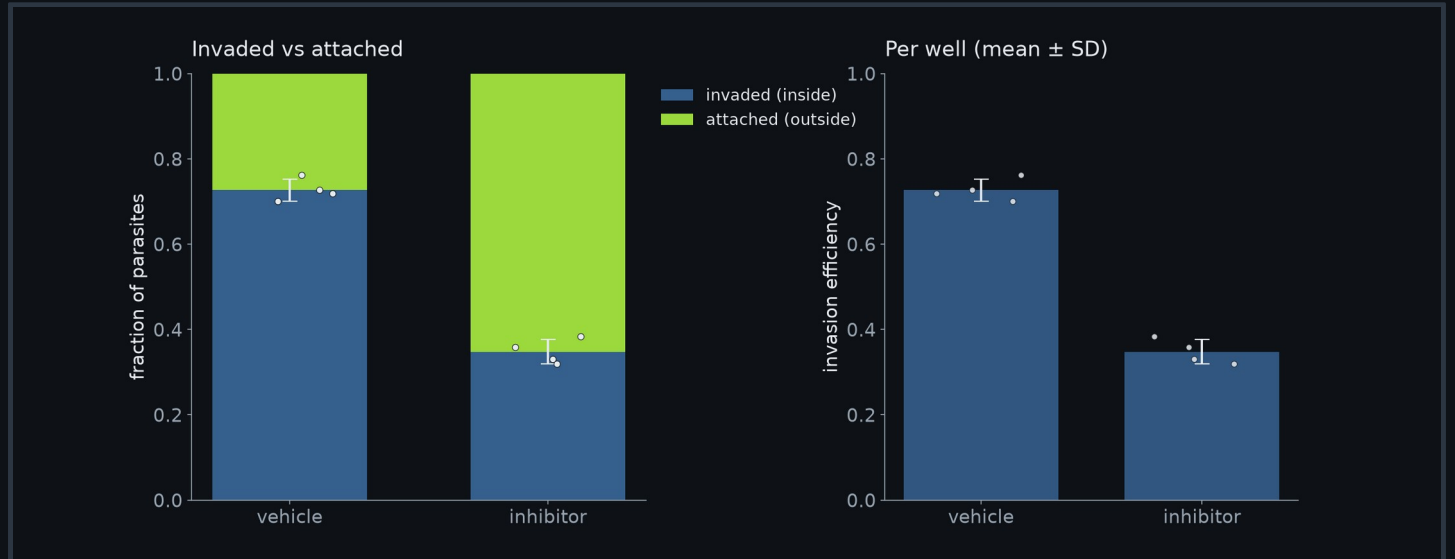
Scores each parasite as outside (attached) or inside (invaded) from two-colour differential staining.

WHAT YOU CAN DO

- Set the outside-stain and total-stain channels
- Thresholds per field, or from control wells
- Flags fields where inside and outside cannot be told apart
- Checks the answer holds if the threshold moves

OUTPUT

- Per-parasite calls, per-field thresholds, per-well invasion
- Statistics comparing conditions, per well
- Plots per well, by condition, and of the thresholds





Replication: parasites per vacuole

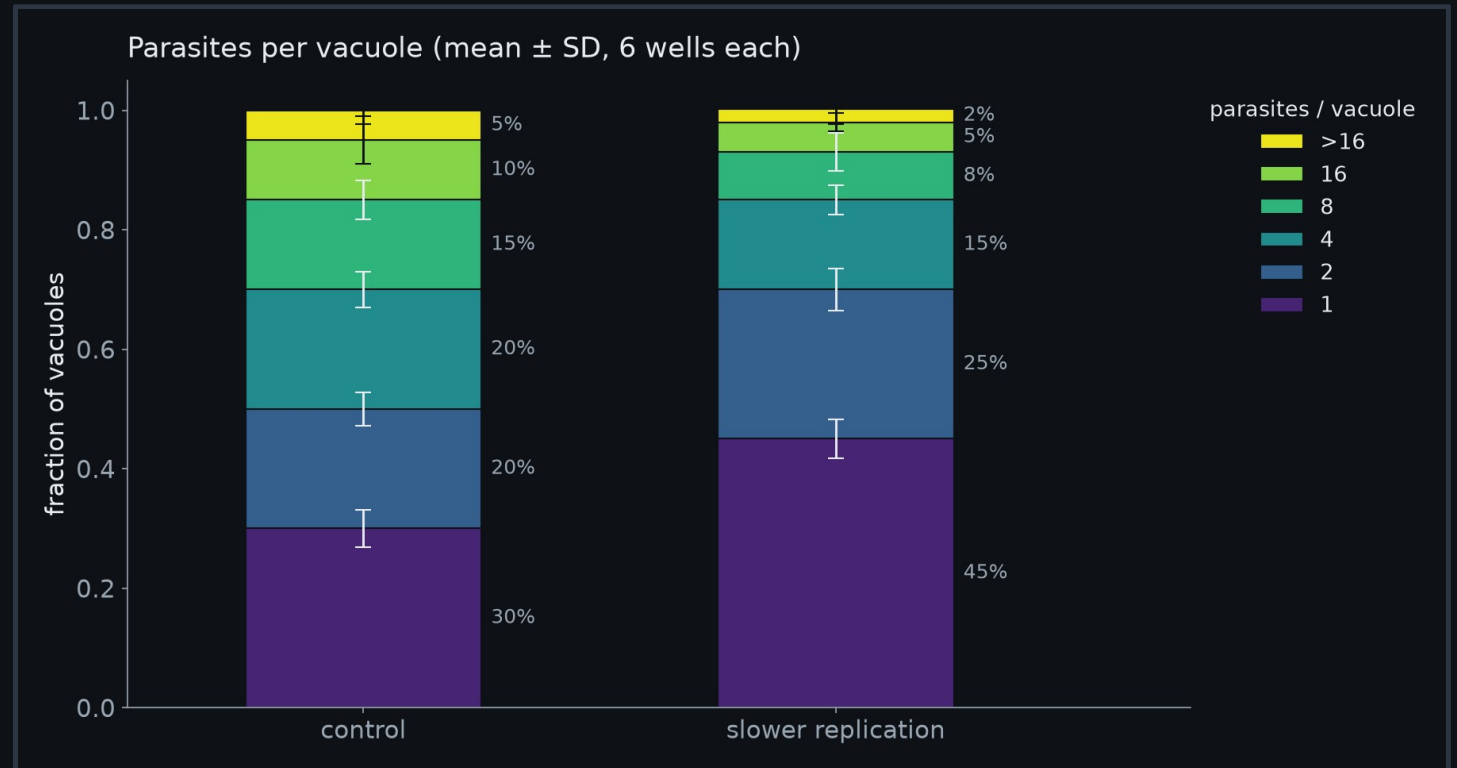
Counts parasites per vacuole (1, 2, 4, 8, 16 ...) and compares the distributions between conditions.

WHAT YOU CAN DO

- Groups parasites into vacuoles: auto, spatial, by cell, or by object
- Area filters for parasites and vacuoles
- Warns when too many vacuoles have a non-power-of-two count

OUTPUT

- Vacuole counts, per-well distributions, condition summary
- Statistics comparing conditions
- Stacked-bar plots per well and by condition



Example output on synthetic data: 6 wells per condition, mean $\pm$ SD

In progress: a classification model that finds and counts Toxoplasma from the Toxoplasma stain, from the nuclear stain, or from the cell mask, so parasites per vacuole can be scored without a dedicated parasite channel.

Project tools in the Help menu



Report

HTML / PDF report of a run: figures, statistics, settings



Pipeline Graph

Which run produced which file, as a graph



Batch Runner

Queue modules with dependencies; validate, then run unattended



Plate Queue

Line up plates for Mask / Measure and let them run overnight



Run History

Every run, its settings and its outputs



Distributed Jobs

Send work to SSH hosts, Slurm or the cloud



Database Browser

Browse, search, filter and export project tables



Data Manager

What is on disk, how big, and what can go



Project Browser

All projects on this machine



Feature Dictionary

What every measured column means, and its unit

05

Trained models

Models trained for Toxoplasma and host-cell images, shipped through the spaCR Model Zoo. Newest version of each.



PV



Plaque



Wells



Cross-channel



MODEL ZOO

Seven models, one line in a settings file

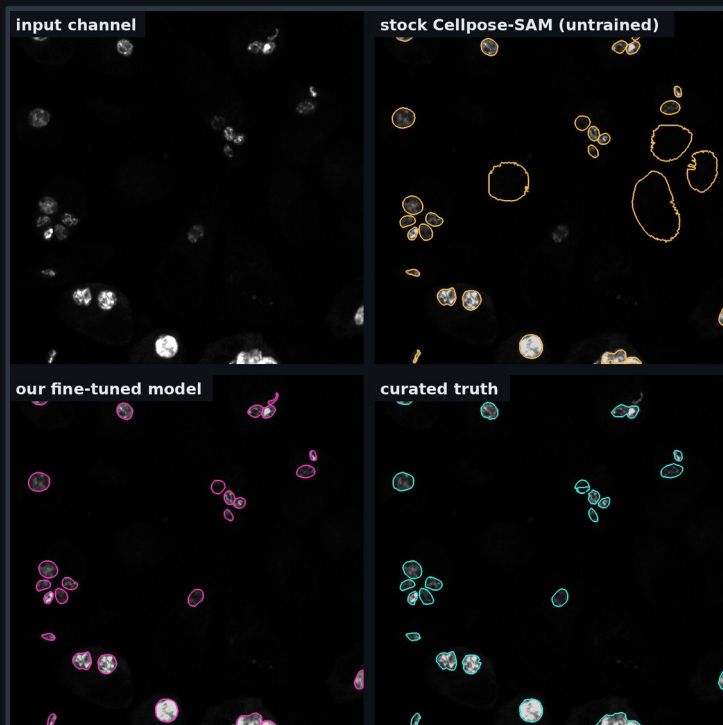
Model	Settings key	Finds	Architecture	Score on unseen images
Toxoplasma PV v3	toxoplasma_pv_v3	Parasite vacuoles	Cellpose-SAM	F1 0.860 (stock 0.765)
Toxoplasma Plaque v2	toxoplasma_plaque_v2	Plaques in wells	Cellpose-SAM	F1 0.819 on published wells
Plaque Well Detector v2	toxoplasma_well_detector_v2	Wells in plates / figures	YOLO26n	mAP50 0.946
Toxoplasma from Cell Mask	toxoplasma_from_cellmask_v1	PVs from the cell channel	Cellpose-SAM	F1 0.606 (stock 0.021)
Nuclei from Cell Mask	nuclei_from_cellmask_v1	Nuclei from the cell channel	Cellpose-SAM	F1 0.888 (stock 0.201)
Cell from Hoechst	cell_from_hoechst_v1	Cells from the nuclear channel	Cellpose-SAM	F1 0.870 (stock 0.301)
Toxoplasma from Hoechst	toxoplasma_from_hoechst_v1	PVs from the nuclear channel	Cellpose-SAM	F1 0.569 (stock 0.002)

How it works Name a key in a settings file: `pathogen_model: toxoplasma_pv_v3:` and spaCR downloads it the first time it is needed. Anyone can contribute a trained model: it is uploaded to the spaCR Hugging Face and then appears in every user's Model Zoo.



MODEL · TOXOPLASMA_PV_V3

Toxoplasma PV v3



Unseen test field. Orange = vanilla Cellpose-SAM (cpsam_v2, never trained on our images), magenta = PV v3, cyan = curated truth

WHAT THE SCORES MEAN

F1 share of objects found correctly; lower if objects are missed or invented (1 = perfect). Found = outlines overlap by at least half.

AJI how well each outline matches; lower if a cell is split in two or two cells merged.

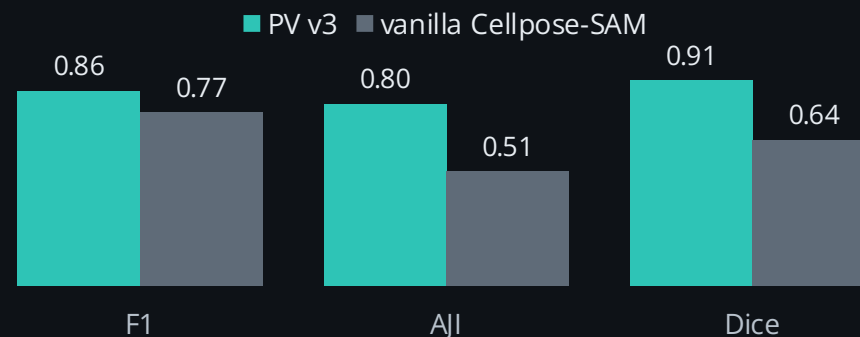
Dice pixel overlap between the model's mask and the true mask.

stock vanilla Cellpose-SAM (cpsam_v2), never trained on our images.

Cellpose-SAM (cpsam_v2), round 6

Outlines Toxoplasma parasitophorous vacuoles in anti-Toxoplasma or DsRed images (RH and ME49).

11 wells never used for training



TRAINED ON

556 hand-curated fields from five rounds of curation: 15,550 vacuoles. 11 wells have never been used for training.

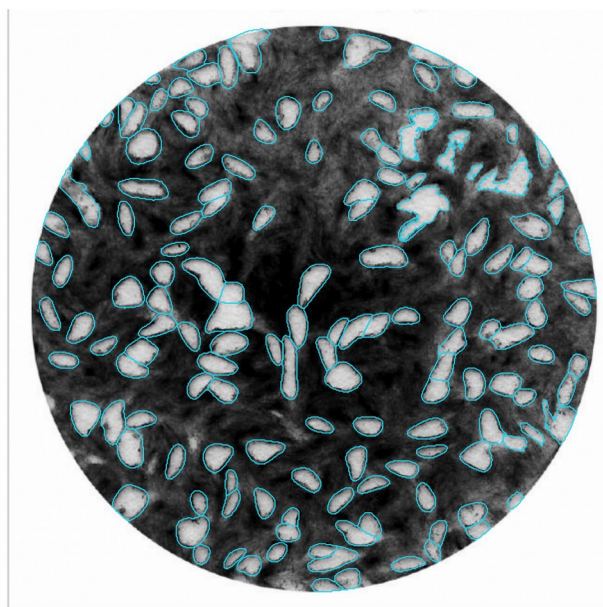
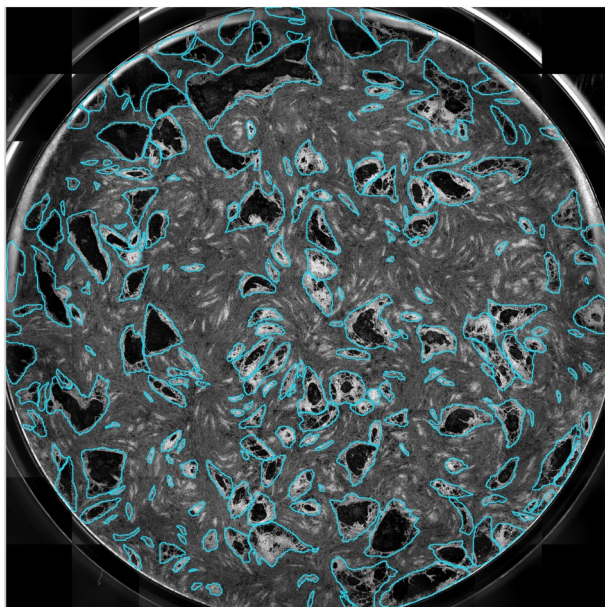
GOOD TO KNOW

- Compared with vanilla Cellpose-SAM, not with PV v2
- Across five different train / test splits: F1 0.82
- Best for counting and size; less exact on fine outline detail
- Scores the same as round 2 on these wells (0.860 vs 0.865)



MODEL · TOXOPLASMA_PLAQUE_V2

Toxoplasma Plaque v2



Curated plaque outlines (cyan): our microscope (left) and a published figure (right)

WHAT THE SCORES MEAN

F1 share of objects found correctly; lower if objects are missed or invented (1 = perfect). Found = outlines overlap by at least half.

AJI how well each outline matches; lower if a cell is split in two or two cells merged.

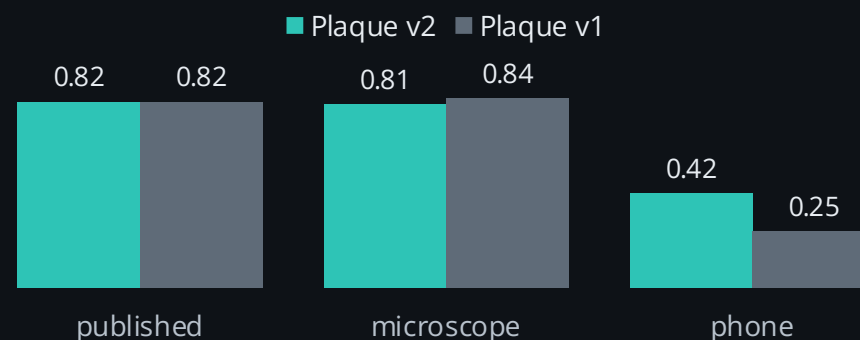
Dice pixel overlap between the model's mask and the true mask.

stock vanilla Cellpose-SAM (cpsam_v2), never trained on our images.

Cellpose-SAM (cpsam_v2), round 5

Outlines plaques in crystal-violet wells: from our microscopes, phone cameras and published figures.

F1 on 81 unseen fields, by image source



TRAINED ON

488 curated fields in four domains: 298 published-figure wells, 96 phone-camera wells, 94 microscope scans: 27,582 plaques.

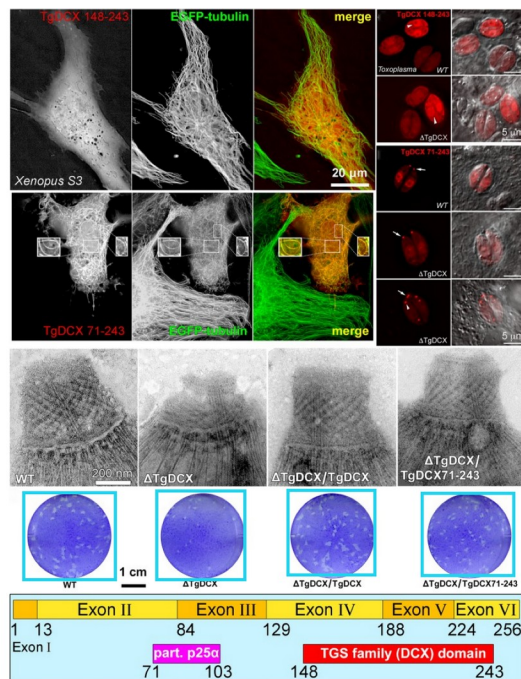
GOOD TO KNOW

- Only plaque model trained on phone images: still misses most plaques there (recall 0.30)
- Balanced precision / recall (0.81 / 0.83); v1 undercounts
- 2 false plaques in 6 blank wells (v1: 19)
- For microscope-only work v1 is still slightly better



MODEL · TOXOPLASMA_WELL_DETECTOR_V2

Plaque Well Detector v2



Detected wells (cyan) in a published figure; fluorescence and EM panels ignored. Shown with the previous detector

WHAT THE SCORES MEAN

precision of the boxes it draws, the share that really are wells.

recall of the real wells, the share it finds.

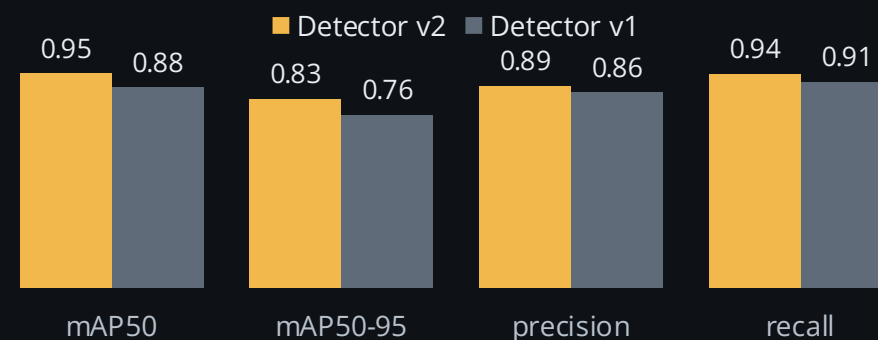
mAP50 overall detection score; a box counts when it overlaps the true well by at least half.

mAP50-95 the same, but also rewards boxes drawn tightly around the well.

YOLO26n object detector

Draws a box around every plaque well in a plate photo or a published figure: the first step before plaque segmentation.

129 unseen images, 84 of them with no well



TRAINED ON

1,070 training images: our plates plus 939 figures from published papers. No paper appears in both training and testing.

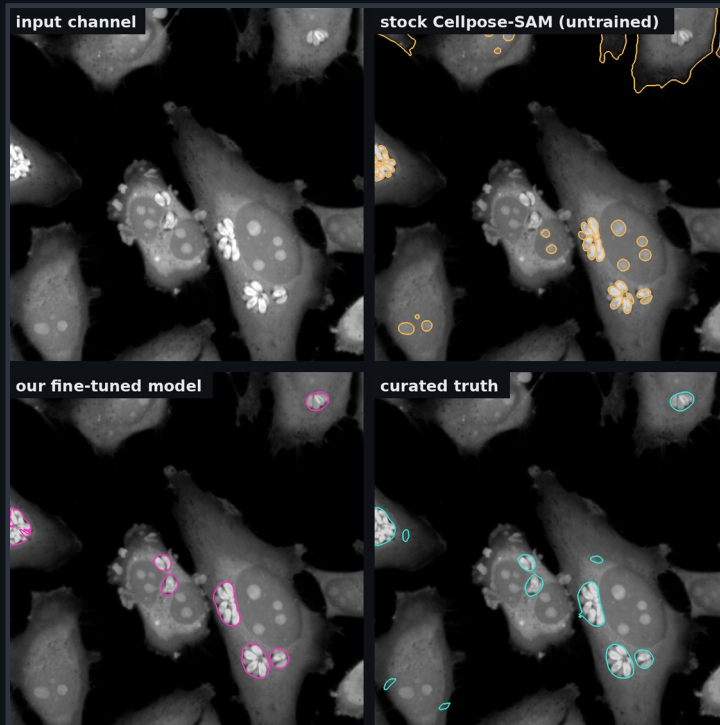
GOOD TO KNOW

- Better than v1 on every score
- False wells on figures with no well: 152 → 49
- Finds wells, not plaques; the well size also gives the mm² scale



MODEL · TOXOPLASMA_FROM_CELLMASK_V1

Toxoplasma from Cell Mask



Unseen test field: orange = vanilla Cellpose-SAM (untrained), magenta = our fine-tuned model, cyan = curated truth

WHAT THE SCORES MEAN

F1 share of objects found correctly; lower if objects are missed or invented (1 = perfect). Found = outlines overlap by at least half.

AJI how well each outline matches; lower if a cell is split in two or two cells merged.

Dice pixel overlap between the model's mask and the true mask.

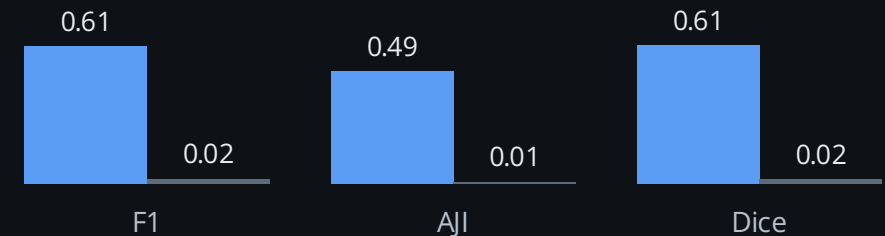
stock vanilla Cellpose-SAM (cpsam_v2), never trained on our images.

Cellpose-SAM (cpsam_v2), cross-channel

Finds Toxoplasma vacuoles from the host cell-mask channel alone: no parasite stain needed.

463 unseen fields

■ fine-tuned ■ vanilla Cellpose-SAM



TRAINED ON

2,567 training fields; tested on 463 fields from other wells. HFF, HeLa and THP-1 hosts.

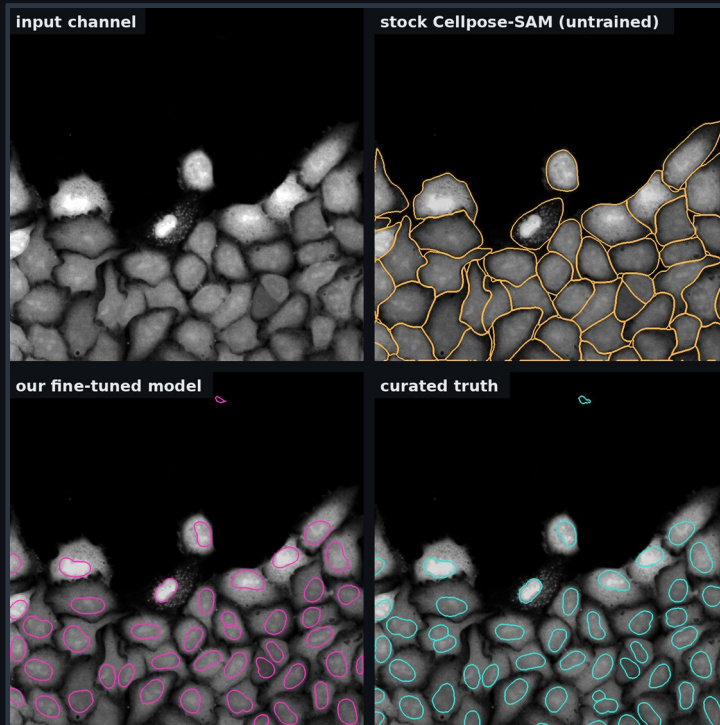
GOOD TO KNOW

- Stock Cellpose cannot do this at all
- F1 by host: HeLa 0.71 · HFF 0.56 · THP-1 0.47
- For counting, occupancy and area: not fine morphology



MODEL · NUCLEI_FROM_CELLMASK_V1

Nuclei from Cell Mask



Unseen test field: orange = vanilla Cellpose-SAM (untrained), magenta = our fine-tuned model, cyan = curated truth

WHAT THE SCORES MEAN

F1 share of objects found correctly; lower if objects are missed or invented (1 = perfect). Found = outlines overlap by at least half.

AJI how well each outline matches; lower if a cell is split in two or two cells merged.

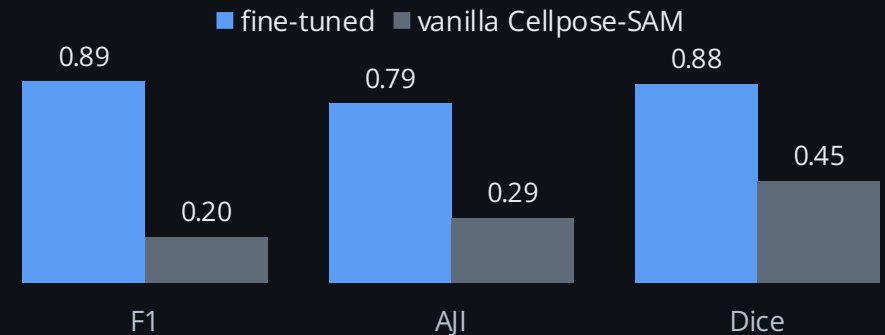
Dice pixel overlap between the model's mask and the true mask.

stock vanilla Cellpose-SAM (cpsam_v2), never trained on our images.

Cellpose-SAM (cpsam_v2), cross-channel

Finds nuclei from the cell-mask channel alone: frees the DAPI / Hoechst channel for another marker.

453 unseen fields



TRAINED ON

Cell-mask images paired with curated nuclear masks; tested on 453 fields from wells never used for training. HFF, HeLa and THP-1.

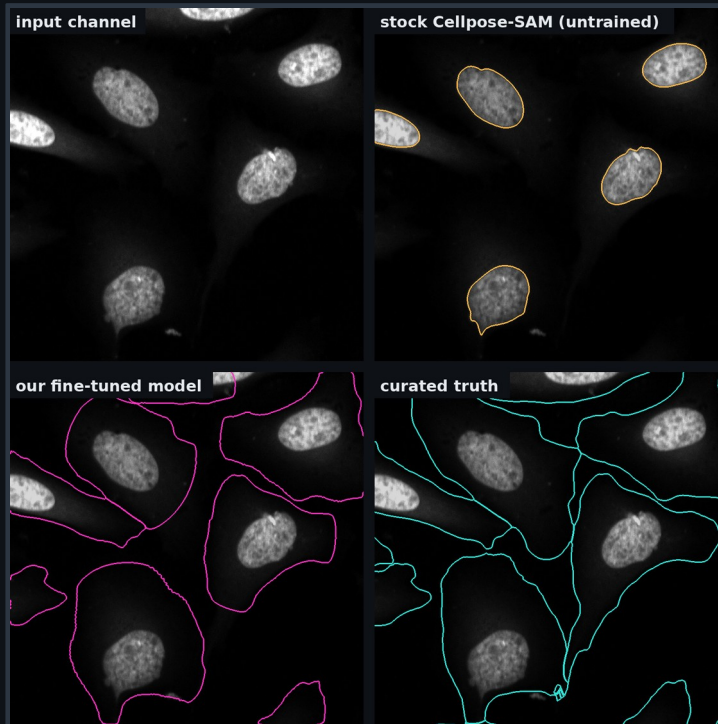
GOOD TO KNOW

- F1 by host: HFF 0.93 · HeLa 0.86 · THP-1 0.86
- Learns nuclei from cell shape: expect worse on unusual or very confluent cells



MODEL · CELL_FROM_HOECHST_V1

Cell from Hoechst



Unseen test field: orange = vanilla Cellpose-SAM (untrained), magenta = our fine-tuned model, cyan = curated truth

WHAT THE SCORES MEAN

F1 share of objects found correctly; lower if objects are missed or invented (1 = perfect). Found = outlines overlap by at least half.

AJI how well each outline matches; lower if a cell is split in two or two cells merged.

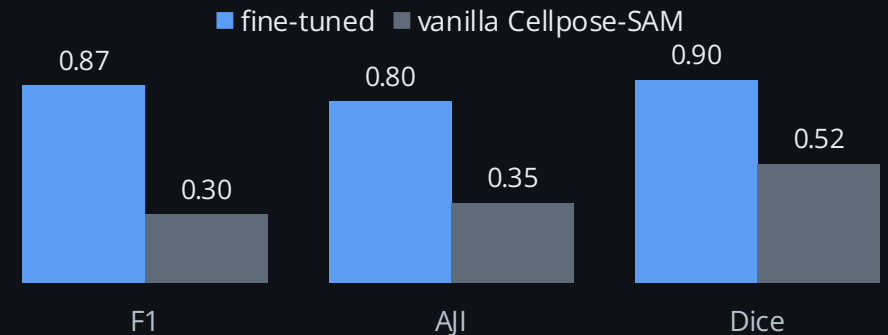
Dice pixel overlap between the model's mask and the true mask.

stock vanilla Cellpose-SAM (cpsam_v2), never trained on our images.

Cellpose-SAM (cpsam_v2), cross-channel

Finds whole host cells from the Hoechst (nuclear) channel alone: the counterpart of nuclei-from-cell-mask.

451 unseen fields



TRAINED ON

2,578 training fields (237,957 cells); tested on 451 fields (45,098 cells) from other wells.

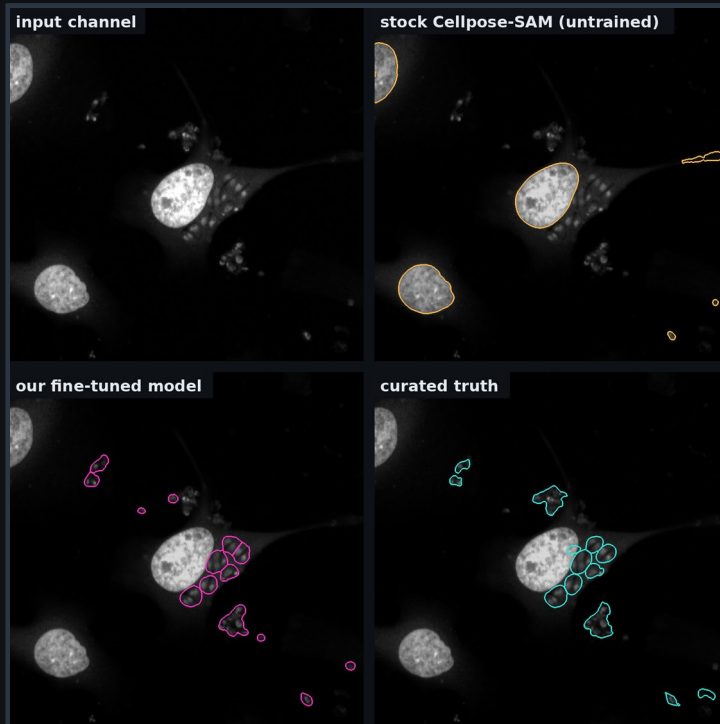
GOOD TO KNOW

- Misses some cells but rarely invents one (precision 0.94, recall 0.81)
- F1 0.57 higher than stock



MODEL · TOXOPLASMA_FROM_HOECHST_V1

Toxoplasma from Hoechst



Unseen test field: orange = vanilla Cellpose-SAM (untrained), magenta = our fine-tuned model, cyan = curated truth

WHAT THE SCORES MEAN

F1 share of objects found correctly; lower if objects are missed or invented (1 = perfect). Found = outlines overlap by at least half.

AJI how well each outline matches; lower if a cell is split in two or two cells merged.

Dice pixel overlap between the model's mask and the true mask.

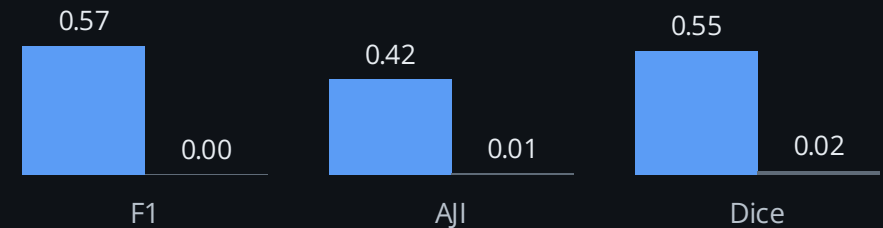
stock vanilla Cellpose-SAM (cpsam_v2), never trained on our images.

Cellpose-SAM (cpsam_v2), cross-channel

Finds Toxoplasma vacuoles from the Hoechst channel alone: no parasite stain needed.

463 unseen fields

■ fine-tuned ■ vanilla Cellpose-SAM



TRAINED ON

2,567 training fields; tested on 463 fields from other wells. HFF, HeLa and THP-1 hosts.

GOOD TO KNOW

- Harder than the cell-mask route (F1 0.57 vs 0.61)
- Stock Cellpose finds essentially nothing
- For counting, occupancy and area



MODEL ZOO

Other models available in spaCR

Segmentation

- Cellpose 4: Cellpose-SAM (default) and Cellpose-DINO
- Cellpose 3 models (cyto3, nuclei, ...)
- SAMCell and DINOCell for label-free / live-cell images
- Cellpose models from bioimage.io
- Models other users have contributed

Image classifiers (Classify)

- ResNet (default ResNet-50), ResNeXt, Wide ResNet
- EfficientNet, MobileNet, RegNet
- ConvNeXt, VGG, DenseNet
- Vision transformers: ViT, Swin, MaxViT
- All start from networks pre-trained on everyday photos

Feature classifiers (Classify)

- XGBoost (default), LightGBM, CatBoost
- Random forest and other tree models
- Logistic regression, support vector machine, small neural network

Embeddings & detection

- Embeddings: ResNet-18 (default), ResNet-50, ConvNeXt, ViT
- YOLO object detection (behind the well detector)

Get started

```
pip install "spacr[qt]" && spacr
```

Docs & tutorials

einarolafsson.github.io/spacr

Source & installers

github.com/EinarOlafsson/spacr

Preprint

[bioRxiv 10.64898/2026.07.08.737057](https://doi.org/10.64898/2026.07.08.737057)

Models

huggingface.co/einarolafsson

Example data

`spacr-download --list`

